



CITY COLLEGE OF CALAMBA

DEPARTMENT OF COMPUTING AND INFORMATICS
Bachelor of Science in Computer Science

**NON-CONTACT OVERLOADING DETECTION SYSTEM
FOR PUBLIC UTILITY BUSES USING YOLO v8
ALGORITHM**

A Thesis
presented to the faculty of Department of Computing and Informatics of the
City College of Calamba
Calamba City

In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

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January 2026



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APPROVAL SHEET



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ABSTRACT

Title: **NON-CONTACT OVERLOADING DETECTION SYSTEM FOR PUBLIC UTILITY BUSES USING YOLOv8 ALGOIRTHM**

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Year: **2026**

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The research titled “Non-Contact Overloading Detection System for Public Utility Buses Using Yolov8 Algorithm” presents an automated approach for monitoring passenger load in public transportation through computer vision. The study explores the integration of real-time image processing and the YOLOv8 object detection model to accurately identify and count passengers without physical contact or manual intervention.

The system utilizes an embedded Raspberry Pi 4, for data processing and integrates essential components such as a camera module and real-time display interface. Furthermore, the study assesses the overall system functionality through testing and validation procedures aligned with Agile development methodology.

Results from system test trials showed detection accuracies ranging from 67% to 82% under varying conditions, with observed limitations caused by occlusion, lighting variations, and crowd density. Real-time processing achieved an average frame processing speed of 0.04–0.09 seconds per frame, supporting timely passenger detection and alert generation. Survey-based evaluation yielded high acceptance, with mean scores of 4.44 for detection accuracy, 4.40 for operational reliability, and 4.38 for alert responsiveness, all interpreted as “Agree” to



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“Strongly Agree.” Evaluation using ISO/IEC 25010 quality standards further indicated strong performance in functionality, reliability, efficiency, security, usability, and maintainability, with most indicators rated above a weighted mean of 4.30.

The findings of this study demonstrate the potential of computer vision-based solutions in assisting transport operators with efficient load monitoring. By providing a non-intrusive and automated method for detecting overloading, the system contributes to improved passenger safety, operational compliance, and technological advancement in public transportation monitoring systems. This research highlights the feasibility of combining modern object detection algorithms with embedded hardware to create practical solutions for real-world transport challenges.

Keywords: *Agile Development, Computer Vision, Embedded Systems, Public Utility Buses, Raspberry Pi, Real-Time Monitoring, YOLOv8*



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DEDICATION

This study is dedicated to the researchers whose dedication, perseverance, and hard work made this study possible, as well as to their families, whose unwavering support, encouragement, and sacrifice played a crucial role in the completion of this journey.



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ACKNOWLEDGMENT

The researchers wish to express their sincere gratitude and appreciation to the following individuals for their support during this study.

The researchers would also like to thank **Regina G. Almonte**, their research coordinator and **Jacqueline A. Dela Torre**, their research adviser. For addressing all their queries throughout the study and for giving insightful advice.

The researchers greatly thank the **panelists and other Computer Science Professionals** for that relevant insight, feedback and advice during those presentations that were of invaluable helpfulness to improve the standards and effectiveness of this study.

The researchers also want to extend their gratitude to the **participants of the study specially to passengers and authorities** for their willingness to share valuable information into the Non-Contact Overloading Detection System for Public Utility Buses that greatly contributed to the research.



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Chapter 1

THE PROBLEM AND ITS BACKGROUND

This chapter presents the problem of overloading in public utility buses, especially in urban environments, and the limitations of existing manual monitoring practices. Overloading still presents safety hazards despite the existence of regulations. This research suggests a non-contact overloading detection system based on the YOLOv8 algorithm to give real-time accurate passenger counts, enhancing enforcement and passenger safety.

Project Context

The Calamba, Laguna transport system continues to face challenges due to the increasing number of vehicles and passengers, with overloading in public utility buses becoming a serious problem. Overloading results in severe safety hazards, shortens the life of buses, and is often the cause of repeated violations of transport policies. At present, overloading is detected through manual observation by traffic enforcers stationed at terminals or checkpoints. These personnel commonly use visual headcounts or physical inspections to estimate the number of passengers inside a bus. This method is frequently time-consuming and prone to human error, reducing its consistency in implementing regulations.

Based on a survey conducted by the researchers, all respondents confirmed that they have not received complaints from the public regarding overloaded buses. However, 83.33% of the respondents believe that the current enforcement methods for detecting overloading are insufficient and ineffective. Additionally, all respondents agree that overloading in public utility buses contributes to increased road accidents and faster wear and tear of the vehicles, making it a serious concern in the country's transport system. Moreover, 100% of the



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respondents expressed support for the use of technology to improve the monitoring of public transportation, with none reporting resistance from PUB operators toward the implementation of stricter systems.

Currently, manual detection by traffic enforcers is the only method of identifying overloaded buses, an approach that is inefficient, subjective, and prone to error. This lack of technological integration creates a gap in regulatory enforcement, often resulting in unchecked overloading, increased road accidents, and reduced vehicle lifespan.

Through this AI-driven method, the system seeks to enhance monitoring effectiveness, provide better data accuracy, and assist in enforcing transport policies more efficiently. It is further intended to interoperate with available systems employed by government and transport operators. It is also able to support efforts towards developing a safer and intelligent public transport in the Philippines.

This project plans to design a non-contact hardware and software system for PUB overloading detection based on the YOLOv8 object detection algorithm. The system will employ a camera device on the rearview mirror to examine video streams and properly tally the number of passengers on board. During overloading, the system automatically sends real-time messages to both the Land Transportation Office (LTO), including of the bus, its location, and time.

Overloading of PUBs remains a persistent issue for urban transport, causing safety hazards, technical breakdowns, and inefficiency in the transport system. While government policies place strict restrictions on passenger capacity, enforcement is challenging due to the absence of an automated monitoring system. This usually leads to continuous violations, making road accidents more likely, passengers uncomfortable, and vehicles damaged earlier



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than expected. In this research, pertinent data will be collected to present tangible evidence of the chronic problem of overloading in Public Utility Buses, in support of the need for an automated detection system.

Research Framework

It is relevant and essential to understand the various factors influencing public utility bus overloading and its impact on transportation safety and efficiency. The theories that can be considered essential in the study are (1) Systems Theory, (2) Human Factors Engineering Theory. These theories provide a strong foundation for developing an automated overloading detection system that enhances regulatory compliance, passenger safety, and public transport efficiency.

System Theory

In the context of public transportation, buses, drivers, passengers, traffic regulations, and safety protocols form a dynamic and interdependent system. Overloading disrupts this balance by increasing wear on vehicles, reducing operational efficiency, and endangering passenger safety. This study applies System Theory by treating the overloading detection mechanism not as an isolated tool but as an integral part of the transportation system. The introduction of a non-contact overloading detection system using YOLO v8 aims to enhance real-time decision-making, improve traffic regulation compliance, and optimize resource allocation such as scheduling, maintenance, and enforcement efforts.

By understanding how each component interacts with the others, the proposed system aligns technical innovation with system-wide improvement, ensuring that changes in one part (e.g., real-time weight monitoring) positively influence others (e.g., road safety, vehicle lifespan, policy enforcement).



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Human Factors Engineering Theory

The Human Factors Engineering Theory emphasizes the design of systems that optimize human well-being and overall system performance. In transportation technology, the usability and accessibility of a system are as important as its technical accuracy.

This study considers the cognitive and physical capabilities of its users bus operators, traffic enforcers, and regulatory bodies. For example, a YOLO v8-based detection system must present information in a clear, actionable format (e.g., visual alerts or dashboard displays) to ensure that decision-makers can respond swiftly and correctly. Moreover, the placement, response time, and feedback mechanism of the system are designed to minimize errors and user resistance.

By integrating Human Factors principles, the system is not only technically proficient but also practical and scalable, increasing adoption and long-term sustainability in real-world transit settings.

Theoretical Framework

The growing concern about passenger safety and the efficiency of public utility buses, addressing the issue of overloading has become more important. Overloading not only puts passengers at risk but also affects vehicle performance and the reliability of public transportation. Because of this, the researchers developed a Non-Contact Overloading Detection System that uses YOLOv8 and GPS technology to detect and monitor overloading in real time without interrupting the operation of buses.

This study applies the Technology Acceptance Model (TAM), a theory that explains how users come to accept and use new technology. The TAM model focuses on two key factors: perceived usefulness and perceived ease of use. In the context of this study, the system



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was designed to provide reliable and timely detection of overload while making sure that users such as government regulators and concerned agencies can operate and access the system without difficulty.

In considering the TAM model in the development process, the system encourages user acceptance and long-term use. It supports decision-making by sending alerts through a mobile application and provides location tracking through GPS. This helps agencies monitor overloading cases more efficiently and take immediate action when necessary. The use of a non-contact system also minimizes disruptions for bus operators while improving road safety for the public. To support both the technical functionality and behavioral adoption of the proposed Non-Contact Overloading Detection System, two key theories are incorporated: Systems Theory and Human Factors Engineering. Systems Theory is a theory that sees any organization or process as a system consisting of connected parts that influence one another.

A change in a single component of the system naturally has an impact on the others, and overall effectiveness depends on how well all of these parts function together. This theory supports the study by recognizing that the transportation ecosystem including buses, regulatory agencies, personnel, and technology is related. The detection system serves as an important component that impacts and enhances the overall efficiency of public transportation.

By automatically identifying and reporting overloading instances using camera feeds and GPS, the system improves coordination between bus operators and enforcement agencies. The distribution of information becomes easier and more responsive, contributing to safe and regulated transportation circumstances. Human Factors Engineering emphasizes building systems that are appropriate for users' physical and mental capacities. This theory strengthens



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the system's usability by ensuring it is accessible to frontline personnel such as transport enforcers. The solution simplifies monitoring and reporting by providing a simple mobile interface, real-time alerts, and clear visual outputs. This allows users to complete their tasks swiftly without having complex technical knowledge, which is critical in fast-paced public transportation operations.

By integrating Systems Theory and Human Factors Engineering, the study creates a strong theoretical foundation that supports both the technical integration of the detection system into the transport network and its practical adoption by end-users.

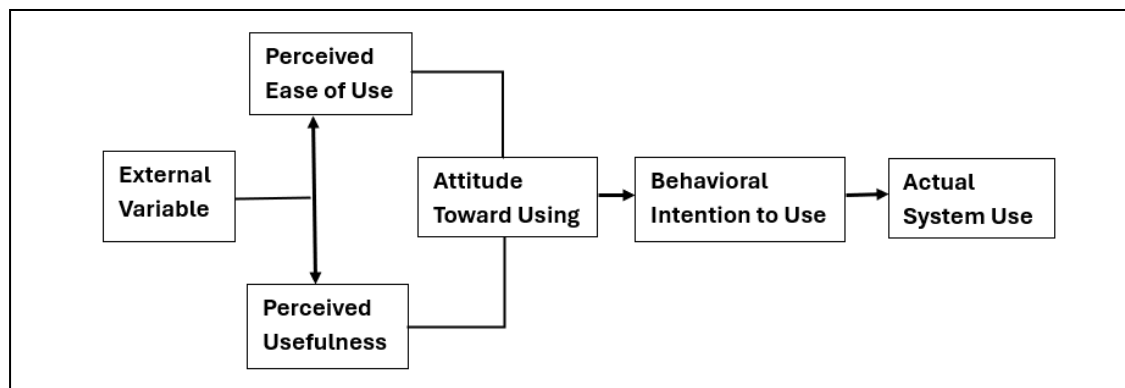


Figure 1-1. Theoretical Framework.

The figure illustrates the application of the Technology Acceptance Model (TAM) to the proposed Non-Contact Overloading Detection System for public utility buses. The external variables are positioned at the top of the model and include theoretical foundations and technical components such as the YOLOv8 algorithm, camera systems, and real-time data processing software. These external factors influence two key user perceptions: perceived usefulness and perceived ease of use. Perceived usefulness refers to the extent to which users—including drivers, transport authorities, and commuters believe that the system enhances transport safety and operational efficiency by accurately detecting passenger overloading without physical contact. Perceived ease of use has to do with how easily the users



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feel that they can engage with or trust the system, which includes feature like automated detection, minimal manual effort, and a simplified interface. The two perceptions guide the overall attitude of the user towards the adoption of the technology, which directly influences their behavioral intention to do so. Lastly, a high desire to apply the system results in real system use in public transport environments, providing safer and more compliant passenger control.

Conceptual Framework

The conceptual framework for the Non-Contact Overloading Detection System covers how the system operates and how its elements interact. This demonstrates how data flows through the system to detect overloading in public utility buses and understanding of the system's design with a focus on how the YOLO v8 algorithm detects and manages bus occupancy.

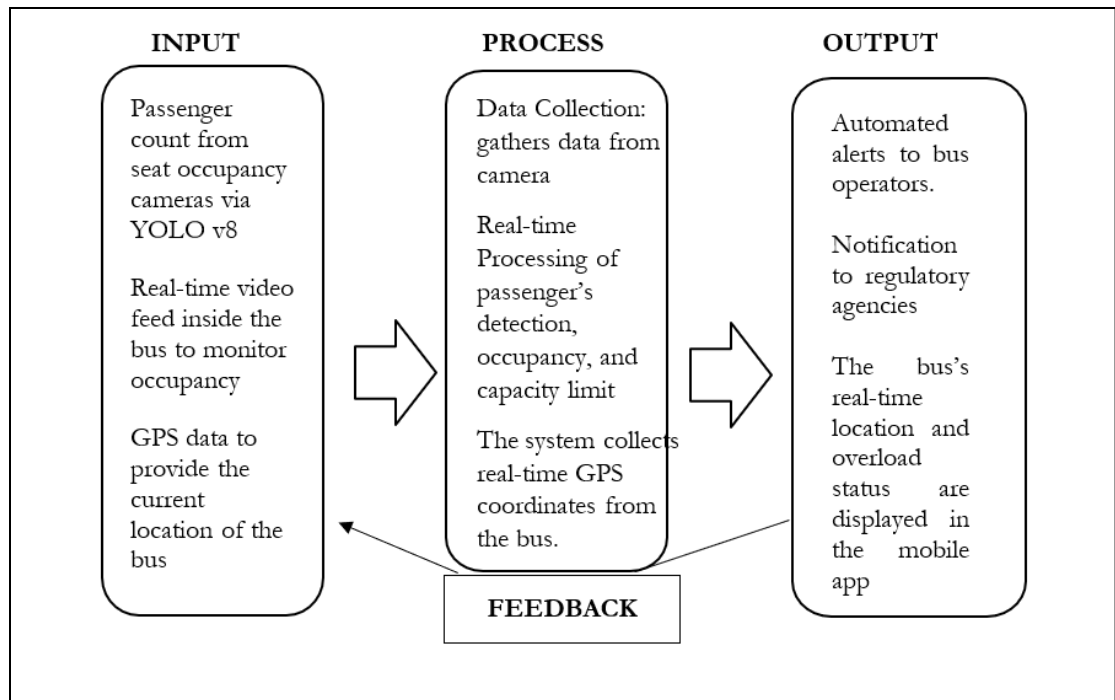


Figure 1-2. Conceptual Framework.



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The input stage contains collecting real-time data from cameras installed inside the bus. The camera captures live video feeds and analyzes using the YOLO v8 object detection algorithm to count the number of passengers. Within the same time, the system also collects location coordinates through GPS modules to determine the exact location of the bus. These inputs are used to assess whether the number of passengers exceeds the allowed capacity, helping ensure compliance with transport regulations.

During the processing stage, once the data is collected, the system processes it in real time. The video feed is analyzed using the YOLO v8 model to detect and count people inside the bus. The system then compares the passenger count to the bus's legal capacity. At the same time, it uses the GPS data to identify where the bus is during each event. If the passenger count goes over the allowed number, the system recognizes the situation overloading and prepares to send a response.

Upon detection of overload, the system provides multiple outputs. It provides alerts to the bus operator via mobile app to notify them immediately. It also inform transportation authorities about the violation so that authorities can take appropriate action. The location of the overloading has been tracked and included in the report, using GPS data. These outputs allow bus operators and agencies to respond quickly and ensure the regulations are followed.

The framework incorporates a feedback loop, allowing the system to adapt and improve based on its performance. This is adhered by refining detection accuracy, optimizing the alert mechanism, and ensuring consistent system reliability across different operational conditions. This approach supports the study's deeper goal of developing a more secure and effective public transportation monitoring system. It minimizes the need for human enforcement, improves response time to violations, and assists in maintaining passenger loads



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within legal and safe limits.

Objectives of the Study

The main aim of this research is to design, deploy, and test a Non-Contact Overloading Detection System for Public Utility Buses based on camera-integrated technologies. The research targets:

1. To create and build a hardware-based system that can detect overloading based on:
 - 1.1. Real-time monitoring through Computer vision.
 - 1.2. Passenger detection through camera and image processing technology
 - 1.3. Data processing and integration through a raspberry pi 4 or embedded system
2. To establish an alert system that automatically notifies concerned stakeholders in case of overloading in terms of:
 - 2.1. Wireless transmission of real-time overloading data to the LTO for instant awareness and action.
 - 2.2. Real-time automatic creation of digital logs and reports for regulatory documentation, monitoring, and enforcement initiatives.
 - 2.3. Installation of a secure, web-based dashboard viewable by authorized LTO staff with the ability to monitor overloading incidents in real time, visualize data, and make simplified enforcement decision-making.
3. To evaluate the system's performance in real-world public transportation settings in terms of:
 - 3.1. Accuracy of passenger detection
 - 3.2. Dependability under different operational conditions
 - 3.3. Responsiveness and promptness of alerts and data transmission



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4. To determine the system's level of effectiveness as evaluated through the ISO 25010 quality standards in terms of
 - 4.1. Functionality
 - 4.2. Reliability and Efficiency
 - 4.3. Security
 - 4.4. Usability
 - 4.5 Maintainability
5. To suggest system enhancements and future development directions with emphasis on:
 - 5.1. Integration with cloud-based monitoring and LTO databases
 - 5.2. Expansion for use in other types of public transport
 - 5.3. Scalability for national-level implementation and long-term monitoring

Significance of Study

The study will be beneficial as it introduces an innovative approach to ensuring compliance with public transportation regulations through the development of the Non-Contact Overloading Detection System. Leveraging computer vision and automated alerts, the system offers a more efficient, accurate, and scalable solution compared to traditional manual monitoring methods. It contributes to the advancement of transportation technologies, improving road safety and operational efficiency. The findings of this study have the potential to inform future policymaking and improve public transportation management on a broader scale.



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Bus Operators

The system provides significant advantages in vehicle maintenance and operational efficiency. By reducing overloading, the technology lowers the damage caused on buses, contributing to cheaper maintenance costs and increased vehicle lifespan. Moreover, the technology helps operators comply with regulations, reducing the possibility of penalties and fines from regulatory agencies. Operators are better able to manage their vehicles and optimize their daily operations.

Passengers

The detection and prevention of overloading using this technology reduces the danger of accidents, discomfort, and longer journey durations. Providing an assurance of security for passengers while also encouraging a more reliable and effective travel experience.

Regulatory Agencies

The system serves as an automated tool to enforce compliance with public transportation regulations. The Non-Contact Overloading Detection System provides a more reliable and accurate means of ensuring that buses operate within legal limits. Therefore, regulatory bodies can focus their efforts on other critical aspects of public safety and law enforcement.

Urban Planners and Policymakers

Leveraging the insights gained from the study, they can make more informed choices regarding the development of sustainable and efficient public transportation systems. The findings contribute to creating policies that can improve traffic management and enhance the overall functionality of urban transit networks.



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Researchers

The results of this study will contribute to the growing body of knowledge on non-contact overloading detection and the application of computer vision in public transportation. Researchers can enhance their system development skills by utilizing the principles and methodologies outlined in this study as a reference for developing more advanced and adaptive transportation monitoring systems.

Future Researchers

Future studies can expand on this work to investigate improvements in detection accuracy, efficiency, and system scalability. Researchers may also investigate incorporating new technologies and machine learning models to improve the Non-Contact Overloading Detection System for more widespread use in public transportation.

Scope and Limitations

This research involves the design and construction of a Non-Contact Overloading Detection System for Public Utility Buses (PUBs), using the YOLOv8 algorithm for real-time object detection. The system aims to track passenger numbers via video feed, allowing automated overloading detection without contact or manual intervention. It seeks to enhance the effectiveness of public transportation by providing an accurate and automated method of tracking and reporting passenger load. The study scope encompasses designing, integrating, and evaluating the performance of the system in selected high-density urban areas where overloading is a chronic problem. The system will also have an automated warning system that sends notices to bus operators and regulatory bodies when the passenger capacity is reached. Performance is to be measured in terms of detection precision, system reliability, and overall operability using the standards and performance criteria applicable to the field.



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However, the study only pertains to public utility buses and does not cover other forms of public transportation like jeepneys, vans, or tricycles. The system exists only as a monitoring and warning device and does not impact or drive boarding behavior, nor is it equipped with enforcement means or penalty systems. Environmental conditions such as lighting, bus movement, and camera placement may affect detection accuracy. The first implementation is also limited to a pilot test across a select number of bus units, which may not accurately reflect the scope of a larger deployment. The research also does not interface with existing transport management or ticketing systems, and its success in real-world applications will be heavily reliant on the adherence of bus operators and the endorsement of regulatory agencies.

Definition of Terms

Automated Alert System. A feature of the detection system that provides real-time notifications to bus operators and regulatory agencies when overloading occurs.

Computer Vision. Advanced sensing devices used to detect passenger load levels in real-time, providing data for automated monitoring and overloading detection.

Human Factors Engineerin. The study of how people interact with technology, ensuring that the overloading detection system is user-friendly and effective for operators, regulators, and passengers.

Non-Contact Overloading Detection System. The proposed system designed to identify and prevent overloading in public utility buses through camera-based monitoring and automated alert mechanisms.

Overloading. The condition where the number of passengers or the total capacity in a public utility bus exceeds the legally allowed capacity, leading to safety risks and operational inefficiencies.



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Public Utility Buses (PUBs). Public transportation vehicles that operate on fixed routes, serving commuters in urban and rural areas.

Real-Time Monitoring. The continuous collection and processing of data to detect overloading as it happens, ensuring immediate corrective actions.

Regulatory Compliance. Adherence to government-mandated passenger capacity limits and safety regulations in public transportation.

Transport Safety. The set of measures and technologies aimed at reducing risks and improving security for passengers in public utility buses.

Urban Public Transport. The system of buses and other public transit services operating within city environments, aimed at providing affordable and accessible commuting options.



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Chapter 2

LITERATURE REVIEW

This literature review discusses the recent developments in intelligent transportation systems and monitoring systems with a focus on the utilization of AI algorithms such as YOLOv8 for object detection in real-time. It also emphasizes machine learning, IoT, and image processing integration to provide applications for traffic management, vehicle detection, crowd density, and public transport optimization. Major issues discussed include autonomous systems, intelligent video surveillance, road safety with fatigue detection, and solutions to urban congestion. The review highlights the increasing importance of embedded systems and deep learning in making today's transportation and operation systems more efficient and reliable.

AI-Based Real-Time Decision Making and Object Detection for Intelligent Transportation and Surveillance Systems

This study aims at the increasing trend of autonomous vehicle interest, development of autonomous systems for efficient and safer road systems has been done. Autonomous vehicle tasks can be categorized into guidance, navigation, and stabilization tasks. Navigation tasks produce road-level routes, while guidance tasks manage autonomous vehicles under complex conditions. Operational-level tasks of stabilization are responsible for transferring tactical decisions to reference trajectories as well as low-level controls. A few methodologies have been suggested for computing effective road-level routes, but navigation is mature technology. Driving policy, or operational-level guidance, is essential for complete autonomous driving. Current driving policies are not sufficient for real-time use in complex



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environments and safety requirements. This paper will contribute to the establishment of a strong driving policy for autonomous cars that can make decisions in real-time in real-time, with highways as a particular transportation network. The method proposed employs Reinforcement Learning (RL) together with a responsibility-sensitive safety model to create RL-based driving policies that try not to cause accidents, drive the vehicle at the desired speed, reduce accelerations/decelerations, reduce lane changes, and maintain passenger comfort.[1]

This study presents a smart video-surveillance system for people detection, counting, and tracking in real time using novel vision processing units (VPUs) hardware modules. The system is based on a low-cost embedded UpSquared2 platform, featuring a Myriad-X VPU with low power consumption and algorithm parallelization capabilities. It incorporates a MobileNet-SSD architecture for people detection and tracking, and a Kalman filter bank for people tracking and counting.[2]

Advanced Vehicle Overload Detection and Real-Time Object Recognition Using YOLOv8 for Smart Traffic Management

This study aims to show that overloading vehicles can negatively impact driving safety, brake system wear, and service life reliability. Current methods for detecting overloaded vehicles are inefficient, leading to static overload detection stations and manual monitoring. A new system based on V2R collects engine speed, torque, and other signals through CAN and DSRC, transmitting data to the roadside detection unit. The system uses automobile dynamics principles to analyze force under driving, calculate vehicle weight, and calculate overload. The system has an accuracy rate of over 85%, making it significant for improving vehicle detection efficiency.[3]



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This study aims to suggest a better YOLOv8 model for enhancing real-time vehicle and pedestrian detection for urban traffic management. The model has a multi-scale feature fusion mechanism, enhanced non-maximum suppression algorithm, and data enhancement. The study was implemented using the VisDrone 2019 dataset, which has many urban traffic videos captured by drones with complex backgrounds and varying illumination conditions. The objective is to offer a high-performance and trustworthy detection solution for urban traffic monitoring systems and encourage intelligent traffic management.[4]

Vehicle Detection and Tracking Using YOLO for Smart Traffic Surveillance in Urban Areas

This study aims to vehicle tracking systems that use computer programs to monitor vehicle location, motion, and status. Detecting complex traffic road video frames is challenging due to background switching. One-stage algorithm methods like YOLO are efficient for detecting vehicles, and other algorithms can be integrated. This research focuses on YOLO's use in vehicle detection and tracking, particularly in traffic management, and its successful implementation in smart cities. It also suggests new model combinations and enhancements in computer vision.[5]

This study aims to increase traffic in major cities and requires smart traffic management systems. This paper discusses an intelligent video surveillance-based car tracking system utilizing neural networks, image-based tracking, and You Only Look Once (YOLOv3). The system can detect, track, and enumerate cars with reasonable outcomes in varying situations, reflecting the necessity for ongoing monitoring and upgradation in metropolis road network management.[6]



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Smart Traffic and Driver Monitoring Systems for Urban Road Safety and Efficiency

This study claims that traffic congestion in urban cities is a serious problem, and real-time information of road traffic density for automobiles is essential for effective traffic control. Causes for congestion are substandard capacity, restrictive demand, and red-light delays. Video surveillance systems and image processing can be used to simulate and optimize traffic signal control. This work shows how one can utilize real-time live video streaming from intersection camera to undertake traffic bulkiness calculation and vehicle detection. The algorithm toggles signal lights according to vehicle density and priority for emergency vehicles, cutting traffic congestion, accidents, fuel usage, and waiting time.[7]

This research study aims to create a driver fatigue detection mobile phone system, which is one of the key causes of car accidents. The system employs the Viola-Jones algorithm for detecting the face and eye blinking of the driver and tracks drowsiness via behavioral signs such as excessive blinking and microsleep. When drowsiness goes beyond a predefined threshold, alarms, loud music, and SMS alerts are issued. The system was able to meet the expectations of the users in terms of functionality, efficiency, portability, reliability, and usability with a 95% accuracy rate for blink rate and 90% for microsleep.[8]

Real-Time Object Detection and Surveillance Systems Using Image Processing and YOLO Algorithm

This research intended to create a "suspicious object detection with alarm notification for security personnel". The primary goal of this research was to monitor or detect the suspicious objects, determine if it was abandoned, and notify the security personnel. A prototype approach was utilized in creating the system which utilized an image processing approach in examining the video footage taken by the camera. The image processing used the



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background extraction, foreground detection, thresholding algorithm, noise reduction algorithm, tracking and detecting algorithm. The system was assessed by 10 security personnel and 10 I.T. specialists using the ISO 25010 criteria in the areas of functional suitability, reliability, performance, efficiency, operability, security, compatibility, maintainability, and portability. The general outcome of the assessment was excellent which meant that the system achieved the standard criteria in system development.[9]

This research study aims to the recognition of objects in real time is a developing technology that carries an enormous possibility for expansion into various technical spheres. Modern object detection systems employing image processing depend on data provided by one frame. A video camera located within the region to be analyzed acquires the picture, observing minutely the dynamics between frames. The You Only Look Once (YOLO) model is an object detection algorithm in images, presently renowned for the precision of data it produces and its quick-working ability. This research outlines a thorough literature review of YOLO studies, and a bibliometric study to chart the trends in the automotive sector from 2020 to 2024. Applications of object detection through YOLO were divided into three main areas: road traffic, development of autonomous vehicles, and industrial environments. A thorough examination was made for each area, with quantitative information on current implementations.[10]

Advancements and Applications of YOLO Algorithm in Real-Time Object and Vehicle Detection

This study aims to show that the YOLO V8 algorithm is a cutting-edge real-time object detection model capable of identifying objects in any image with multiple classes. It's trained on a high-end GPU, which shortens training time and character identification time. The model



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has a 95% recall and a 99.2% precision to identify college buses under any weather condition. Its mAP50 measure of 99% indicates that it possesses a high level of confidence in the identification of items. This solution reduces human effort and time, thus making it a best solution for vehicle entering system automation.[11]

This research paper analyzes the YOLO family of deep-learning object detection algorithms, focusing on their enhanced detection speed and accuracy. It analyzes their industrial application fields, usage examples, and optimization methods. The paper also performs typical tests for performance on various data sets and indicates prospects and difficulties for the development directions of YOLO series algorithms.[12]

Smart Transportation Systems Using IoT and Deep Learning for Real-Time Traffic and Crowd Density Management

The study aims to prove that the Internet of Things (IoT) is an important technological revolution that links different smart devices so they can communicate and share data seamlessly. IoT technology is most beneficial in transport, where it provides advantages like traffic control, enhanced logistics, effective parking systems, and greater safety protocols. Other technologies such as machine learning, big data, and distributed ledgers are under investigation to enhance the advantages of smart transport further. This paper discusses the different technologies employed in smart transportation and their corresponding challenges, including communication mechanisms, architectures, and frameworks. It also discusses communication protocols such as Wi-Fi, Bluetooth, and cellular networks, cloud computing, edge computing, and fog computing. The paper also touches on current challenges in the smart transportation sector, including data privacy and security concerns, network scalability, and IoT device interoperability.[13]



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This research study aims to show that the population increase in Turkey has contributed to high levels of congestion in transportation networks. Congestion translates into delays for people using public transport, resulting in them being forced to reach their destinations early by hours, thereby wasting a lot of time. Passengers do not always have access to up-to-date information regarding the present capacity of public transport vehicles, and hence uncertainty and possible inefficiencies while waiting at bus stops. In other situations, public transport vehicles get fully occupied, with people at the stop going unserved. Additionally, passengers must also know the crowd composition at each stop, since this affects their ability to board and travel efficiently, especially on smaller vehicles like minibuses where stop density is an important consideration. Consequently, passengers need to make decisions based on vehicle and stop density as well as the predicted vehicle arrival time. In order to counter these challenges, the researchers suggest a deep learning-based mobile platform intended to identify crowd density at bus stops and in public transport vehicles. The platform will alert both the transit manager and the passengers regarding real-time density information.[14]

Optimizing Vehicle and Small Object Detection Using Deep Learning in ITS

This study aims to show that the growing need for motor vehicle detection using artificial intelligence in Intelligent Transportation Systems (ITS) applications is the key to unlocking the importance of developments in this area. The release of YOLOv8, the newest in the series of YOLO algorithms, opens a new direction in investigating the capability of this detection algorithm in the ITS sector. The algorithm has never been tested on tasks like vehicle detection before, which presents an opportunity to explore its strengths and contributions in vehicle detection and traffic monitoring. The purpose of this study is to fill this gap by using



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YOLOv8 for the detection of vehicles in the overall context of ITS applications. There was a lot of testing conducted through datasets obtained via aerial observation with a drone. Care was taken to provide a diverse range of target objects in dataset generation, a factor often overlooked in similar studies. Training of the algorithm not only assessed its data handling and generalization but also indicated its next-time potential. The model was put through an exhaustive set of performance tests, both identifying strengths and weaknesses and delineating capabilities and limitations. In comparative analysis, the research compared YOLOv5 and YOLOv8 performance metrics systematically. The results of the study underscore the weakness of the detection process, blaming misclassifications on issues like variations of vehicle shapes, lighting, and relative sizes.[15]

This research study aims at extreme small-scale object detection, a process that uses computer vision and machine learning techniques to identify small objects, which can be difficult to detect due to limited sensor resolution and low contrast. Researchers have developed specialized algorithms using techniques like super-resolution, multi-scale processing, and deep learning to improve object detection accuracy. Two-stage algorithms, such as Faster R-CNN and RetinaNet, generate potential object regions in images and use a classifier like a convolutional neural network (CNN) to determine if each region contains an object of interest. One-stage algorithms, like YOLO and SD, use a single neural network to generate region proposals and perform object classification. These algorithms are suitable for real-time applications and can be trained using a smaller dataset but may not perform as well in terms of accuracy and detection of small objects.[16]



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Application of YOLOv8 in Real-Time Urban Risk and Vehicle Detection for Intelligent Transportation Systems

This paper presents an approach to determining levels of urban flood risk through the submerged condition of buses observed from social media photographs. 1008 image complex scene dataset and labeling using Labeling is constructed. Four different configurations of YOLOv8 are verified in detecting the levels of urban flood risk. Experiments reveal buses to show more stability and precision in detection than using customary reference objects. The approach offers a secure basis for real-time flood hazard assessment and emergency management of urban transport systems.[17]

This research study aims to delve into real-time vehicle detection with the YOLOv8 algorithm, namely YOLOv8n, the most compact scale model in the YOLOv8 series. The ITS-optimized model has a high detection accuracy rate of over 80%, and it is a powerful object detector for real-time smart traffic management. The research points out the potential of YOLOv8 to promote object detection technology in transportation research.[18]

Optimization and Automation of Operational Processes through Simulation and Smart Technologies

This research was on the determination and definition of the process for purposes of maximizing the allocation of workers to job orders sent by the manufacturing service centers of Biñan, Calamba and Sta. Rosa. Using a quantitative approach to research, the research established the wants and needs of passenger vehicle owners based on 399 respondents from areas selected. This formed the basis of the nature of business to be operated. Further, applying qualitative technique, the usual processing time to fulfill job orders was also found out. It will aid in lessening the time spent to process the job order as well as deciding what can improve



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in comparison with its rivals. Utilization of simulation as a remedy is the proposal put forth by researchers that helps instantly conclude how and when the job will be executed sooner. Considering the job's weight or the extent of damage will also assist in the identification of the critical factors of the processes.[19]

This research aims to enhance contact tracing for COVID-19, resolving problems such as manual tracing, accuracy, and user privacy. The researchers created an automated application named WeACT, employing QR codes and GPS technology to trace and alert users of close contact with COVID-19-positive patients. The application shows user data and creates distinct QR codes. The prototype was rated highly by 50 users and 20 experts, saving time spent on tracing contacts. WeACT, based on the study, offers a convenient method of gathering information required for contact tracing.[20]

Synthesis

Overloading in public transportation remains a pressing issue, particularly in densely populated urban areas, leading to safety hazards and operational inefficiencies. Traditional methods of monitoring passenger load often rely on manual counts, which can be intrusive, bias and prone to inaccuracies. Recent advancements in computer vision and deep learning have paved the way for non-contact, automated solutions to address this challenge.

The YOLO (You Only Look Once) family of algorithms has been at the forefront of real-time object detection. The latest iteration, YOLOv8, offers enhanced accuracy and speed, making it suitable for applications in intelligent transportation systems. For instance, Prastyo et al. (2024) developed an automatic passenger counting system for public buses using YOLOv8, achieving high accuracy rates in real-time detection scenarios. Similarly, Budiharto



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and Ngarianto (2024) utilized YOLOv8 to detect and count vehicles on toll roads, demonstrating its efficacy in traffic congestion management.

However, challenges persist. Environmental factors such as lighting variations, occlusions, and camera angles can affect detection accuracy. Moreover, while YOLOv8 provides robust detection capabilities, its performance can be further enhanced by incorporating attention mechanisms and optimized loss functions, as explored by Guo et al. (2024) in their improved YOLOv8 model for vehicle detection.

Despite these advancements, a notable gap exists in the application of these technologies for real-time enforcement of overloading violations in public transportation. Most studies focus on detection and counting, with limited exploration into integrating these systems with enforcement mechanisms. Addressing this gap, our study proposes a comprehensive system that not only detects overloading in PUBs using YOLOv8 but also integrates with enforcement protocols to ensure compliance with passenger limits. By bridging the detection enforcement divide, this research aims to enhance passenger safety and operational efficiency in public transportation systems.



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Chapter 3

RESEARCH METHODOLOGY

This chapter presents the methodology used in the development of the proposed system. It includes the research locale, target population, data analysis plan, software development methodology, and system overview. Furthermore, it identifies the software and hardware requirements and where they are situated within the development phases. Visual representations such as diagrams and system layout are also included to support the technical aspects of the study.

Research Design

This study will create a hardware and software system that can sense passenger overloading in public utility buses (PUBs). The system will employ a camera device placed on the front view to correctly tally the number of passengers. Upon overload, the system will instantly notify the bus operator and the Land Transportation Office (LTO) with real-time information such as of the bus, location, and timestamp. This study employs a quantitative research design in measuring the efficiency of the Non-Contact Overloading Detection System for Public Utility Buses. The study largely employs a descriptive quantitative approach, where systematic measurement and analysis of how efficiently the system detects overloading with computer vision are undertaken. The research aims to measure critical variables such as accuracy, efficiency, response time, and enhancement of compliance among public transport.

Besides, a developmental research methodology is integrated to document the system design, development, testing, and usability. This ensures that the system is traced systematically from conceptualization to deployment, measuring its practicability and effectiveness in real-



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world applications. By integrating these methodologies, the research provides evidence-based data on the effectiveness of the system and potential improvements for future transportation monitoring systems.

Research Locale

This research adopts a descriptive research design to test the efficacy of the Non-Contact Overloading Detection System for Public Utility Buses. The study is mainly descriptive quantitative in nature, enabling systematic measurement and analysis of the performance of the system in detecting overloading using computer vision. The research seeks to measure major variables that include accuracy, efficiency, response time, and improvements in compliance in public transportation.



Figure 3-1. Research Locale.

Map showing the South Luzon Expressway (SLEX) and National Highway routes through Cabuyao and Calamba, Laguna. Key SLEX exits are identified, including Cabuyao, Silangan, Mayapa, Batino, and Calamba. The map highlights the strategic connectivity to surrounding urban and industrial areas.



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Population of the Study

This study involved two primary groups of respondents, namely passengers and transport authorities, who are directly affected by and involved in public utility bus operations. These groups were selected to provide comprehensive feedback on the effectiveness, reliability, and practicality of the proposed Non-Contact Overloading Detection System. The passenger group represents the primary users of public transportation, while the authorities represent regulatory and enforcement personnel responsible for monitoring and ensuring compliance with passenger capacity regulations. The total population and corresponding sample sizes for each group are presented in

Table 3-1. Population of the Study.

SECTOR/AGENCIES	TOTAL OF POPULATION	SAMPLE SIZE USED	COMPUTED SAMPLE SIZE
Passengers	50	44	33
Authorities	10	6	9

Based on initial field observations conducted at the Calamba terminal servicing routes to Manila, it was found that each public utility bus accommodates up to 44 passengers. On average, 6 to 8 buses operate daily along this route, completing approximately 24 trips per day. Although an official count of daily commuters is not available due to the absence of recorded data, an estimated population figure was derived based on these observations. Assuming that 6 buses operate at full capacity over 24 trips, the system potentially accommodates 6,336 passenger slots per day ($44 \text{ passengers} \times 6 \text{ buses} \times 24 \text{ trips}$).

Given that the study seeks to evaluate the Non-Contact Overloading Detection System, the researchers chose to concentrate on individuals with direct experience and knowledge of transportation operations and enforcement. These include bus operators and



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regulatory authorities, as they are the key stakeholders involved in monitoring and managing overloading incidents in public utility buses.

To determine the minimum recommended number of respondents for this study, Yamane's Formula (1967) was used. This formula is commonly applied in survey research involving a known population size, as it provides a statistically appropriate sample while maintaining a specific level of precision.

Yamane's Formula:

To determine the appropriate sample size for the study, Yamane's Formula was utilized. This formula provides a simplified method for calculating sample size based on the total population and a specified margin of error. It ensures that the selected respondents adequately represent the entire population while maintaining the desired confidence level.

Yamane's Formula is used to determine the appropriate sample size for a study based on the total population and the desired margin of error. The formula is expressed as $n = \frac{N}{1 + N(e)^2}$, where n represents the required sample size, N is total population size, and e is the margin of error expressed as a decimal (for example, 0.05 for 5% or 0.10 for 10%). This formula provides a simplified and reliable method for computing the minimum number of respondents needed to ensure that the sample adequately represents the population while maintaining the desired level of precision and confidence. The variables used in Yamane's Formula are defined as follows: n = sample size, N = population size, e = margin of error (set at 0.10 for a 90 % confidence level)

Using an estimated total population of 50 individuals (Passengers = 50) and (Authorities = 10), the sample size computation is as follows:



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Computation of Passengers:

To determine the sample size for passengers, Yamane's Formula was applied using the total population of 50 and a margin of error of 0.10. By substituting these values into the formula, the computed sample size 33 respondents.

Computation of Authorities:

The sample size for authorities was computed using Yamane's Formula with a total population of 10 and a margin of error of 0.10. The computation resulted in a required sample size of 9 respondents.

Although the computed sample sizes recommend 33 passengers and 9 authorities, the researchers also considered practicality, accessibility, and participant's willingness to respond. As a result, the study obtained 44 passenger respondents and 6 authority respondents.

While the computed sample sizes recommended a total of 33 passengers and 9 authorities, the researchers considered practicality, accessibility, and willingness of participants. As a result, the study obtained 44 passenger respondents and 6 authority respondents.

In conducting this research on the Non-Contact Overloading Detection System for Public Utility Buses, the researchers aimed to gather insights from key stakeholders who have firsthand knowledge of bus operations and regulatory enforcement specifically bus operators and regulatory officers. These individuals were selected as the target population due to their direct involvement in managing and addressing issues related to overloading in public transportation. To determine a statistically appropriate and manageable number of respondents, the researchers employed Yamane's formula. This method is widely used in



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survey research for its simplicity and effectiveness in calculating sample sizes from a known population while maintaining a specified level of precision.

Sampling Design

This study employed purposive sampling, a non-probability sampling method used to select participants based on their relevance, expertise, or direct involvement in the subject of investigation. This technique is appropriate for studies that require input from individuals who possess specific knowledge or experience regarding a system or issue. In this research, respondents were deliberately chosen based on their direct involvement in public transportation operations and regulatory enforcement. These include: **Passengers** who regularly use public utility buses Authorities such as regulatory officers and transport personnel responsible for monitoring and enforcing transportation policies

A total of 50 respondents were included in the study, consisting of 44 passengers and 6 authorities. Although the computed sample sizes recommended a minimum of 42 respondents (33 passengers and 9 authorities), the final sample depended on participant availability and willingness. Nevertheless, the number of collected responses remains sufficient to provide meaningful insights into the evaluation of the proposed Non-Contact Overloading Detection System.

Data Gathering Tools

To ensure the collection of reliable and comprehensive data for this study, the researchers will employ a combination of data gathering tools, including survey questionnaires, interviews, and field observations. These tools will be used to collect both qualitative and quantitative data, providing a well-rounded perspective on the effectiveness and operational aspects of the Non-Contact Overloading Detection System for Public Utility Buses.



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Survey/Questionnaires: Survey questionnaires will be administered to passengers, and authorities to evaluate the Non-Contact Overloading Detection System. These surveys will use predefined indicators and multiple-choice questions to ensure consistent and analyzable data. Aligned with the ISO/IEC 25010 quality model, the questionnaire will assess key attributes such as functionality, usability, performance efficiency, compatibility, reliability, maintainability, security, and user satisfaction. This will provide a structured and comprehensive evaluation of the system's effectiveness, ease of use, and integration into existing transport operations, helping to identify areas for improvement and support its potential wider adoption.

Interviews: In-depth interviews will be conducted with major stakeholders, such as bus operators, transportation authorities, and technical experts, to gain qualitative insights into the system's effectiveness. These interviews will focus on understanding the practical implications of the overloading detection system on regulatory enforcement and transportation safety. The interview questions will be designed to assess the impact of individual system components, such as the performance of the YOLO v8 algorithm in detecting overloading, the accuracy of the detection results, and the feasibility of implementing the system on a wider scale. The data collected through interviews will help refine the system and guide future improvements.

Library Research: The researchers reviewed academic theses, journals, and published research materials from various institutional libraries, including those available through online platforms such as the Philippine E-Journals (PEJ), University of the Philippines Digital Archives, and De La Salle University's Animo Repository. These sources provided valuable local context regarding transportation challenges in urban areas, previous research on



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overloading detection systems, and the application of AI in regulatory compliance. Printed materials, including engineering books, transport regulation manuals, and information systems development guides, were also reviewed to gain insights into hardware integration and system architecture design.

Web Browsing: Web-based research supplemented library findings with up-to-date global insights on technologies and methodologies relevant to the project. The researchers accessed reputable sources such as IEEE Xplore, ScienceDirect, Google Scholar, and ResearchGate to explore studies involving the YOLO (You Only Look Once) object detection algorithm, smart traffic systems, and real-time surveillance solutions. International studies focusing on vehicle detection, crowd density estimation, and smart city applications were particularly relevant to the conceptualization and technical implementation of the system.

In addition, the researchers reviewed government and transportation agency websites (e.g., Land Transportation Office [LTO], Department of Transportation [DOTr]) to align the system's features with local policies, guidelines, and compliance requirements.

Field Observations: Field observations will be carried out by deploying the detection system in operational environments to monitor its real-time performance. The researchers will observe how well the system detects overloading in different bus conditions, including varying passenger loads, lighting conditions, and traffic scenarios. This observational data will allow the researchers to identify potential system limitations and refine the algorithm for improved accuracy. The field observations will also offer insights into any operational challenges that may arise during the implementation of the detection system, ensuring that it is both practical and efficient in real-world settings.



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Data Gathering Procedure

The data collection process will be done in a systematic manner to ensure the quality and reliability of collected data. As the first step, a preliminary study will be undertaken to investigate current overloading monitoring systems and determine areas where the current system is lacking. Second, the Non-Contact Overloading Detection System will be designed and implemented for some selected public utility buses and tested on a pilot basis. In this phase, the system will gather real-time data through computer vision, and researchers will send survey questionnaires to bus operators and regulatory officials to obtain their opinions. At the same time, system logs will be captured to track accuracy and response times. Interviews with regulatory agencies will also be carried out to obtain additional information on policy enforcement and compliance. Upon data collection, the collected information will be interpreted using suitable statistical techniques, and the findings will be interpreted to assess how effective the system is. Finally, depending on the findings, recommendations for how it can improve and further strengthen in the future will be formulated.

Data Analysis Plan

The information gathered through surveys, system logs, and interviews will be subjected to statistical and comparative analysis for objective assessment. Descriptive statistics will be employed for summarizing responses from surveys and system performance data to present an overview of accuracy, efficiency, and usability. Accuracy and efficiency measures will be employed for gauging the system's effectiveness in identifying overloading events correctly and responding to different passenger loads. Comparative analysis will be used to contrast the system's output with conventional manual monitoring approaches to ascertain improvements in overloading detection and compliance with regulations. Thematic analysis



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will also be used to make sense of qualitative data collected through interviews, yielding important insights on policy enforcement, user experience, and possible system improvements.

Five-Point Likert Scale

The Likert scale will serve as a primary measurement tool for collecting evaluative feedback regarding the performance, reliability, and overall effectiveness of the Non-Contact Overloading Detection System for Public Utility Buses. This psychometric scale gathers perceptions and attitudes by prompting respondents to indicate their level of agreement with specific statements, ranging from “Strongly Disagree” to “Strongly Agree.” Through this method, the study can translate qualitative insights into quantifiable data that reflect the stakeholders’ assessment of the system.

In this research, the 5-point Likert scale will be used to evaluate how experts, system testers, and relevant stakeholders perceive various aspects of the system, based on its accuracy, usability, efficiency, and functional performance. Each numerical rating corresponds to a defined interval and interpretation, ensuring consistency in the evaluation process:

Table 3-2. Five-Point Likert Scale.

LIKERT-SCALE DESCRIPTION	LIKERT-SCALE	LIKERT-SCALE INTERVAL
Strongly Disagree	1	1.00 – 1.80
Disagree	2	1.81 – 2.60
Neutral	3	2.61 – 3.40
Agree	4	3.41 – 4.20
Strongly Agree	5	4.21 – 5.00

By utilizing this five-point scale, the study allows respondents to express both positive and negative feedback with the flexibility of a neutral midpoint. This is particularly important when evaluating complex system behaviors such as passenger detection accuracy, real-time



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monitoring responsiveness, system reliability under varying lighting or crowd conditions, and the overall usability of the interface and data-processing workflow.

The inclusion of a neutral option ensures that respondents who are undecided or have balanced opinions can provide an accurate representation of their perspective without being forced into agreement or disagreement. This helps maintain the validity of the data and prevents biased or forced responses.

Mean

The mean was employed as a statistical measure to analyze the responses gathered from the survey, which utilized a four-point Likert scale. This scale ranged from strongly disagree to strongly agree, allowing participants to express varying levels of agreement with the survey items. By calculating the mean, the researchers could summarize the overall trends in the data and evaluate whether the responses reflected positive outcomes.

The formula used to compute the mean is, employed as a statistical measure to analyze the responses gathered from the survey, which utilized a four-point Likert scale ranging from strongly disagree to strongly agree. This scale allowed participants to express varying levels of agreement with the survey items. To compute the mean, the researchers used the formula $\bar{x} = \Sigma x_i / n$, where Σx_i represents the sum of all individual scores in the dataset and n refers to the total number of responses. By applying this formula to the survey data, the researchers derived an average score that served as a quantitative indicator of the overall sentiment of the participants. A mean score approaching the higher end of the four-point Likert scale (for example, closer to 3.5 or 4) was interpreted as a sign of positive outcomes, suggesting that the system performs effectively based on the respondents' evaluation.



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By applying this formula to the survey data, the researchers will derive an average score that will serve as a quantitative indicator of the overall sentiment of the participants. If the mean score approaches the higher end of the four-point Likert scale (e.g., closer to 3.5 or 4), it will be interpreted as a sign of positive outcomes, suggesting that the system performs effectively and meets the expectations of its intended users. On the other hand, lower mean scores will indicate less favorable perceptions, highlighting areas that may require further refinement.

This analytical approach will allow the researchers to comprehensively evaluate the impact of the system and determine whether it meets its intended objectives particularly in terms of usability, functionality, efficiency, and reliability as defined by ISO 25010. The results gathered from this process will provide valuable insight into how the participants perceive and experience the system in real-world scenarios.

Performance Metrics

In the context of this study, the use of the pre-trained YOLOv8 model plays a vital role in accurately detecting and counting passengers inside public utility buses. To evaluate the system's detection performance, object detection metrics such as Mean Average Precision (mAP), Precision, and Recall were used. These metrics serve as a standard for assessing how accurately the system identifies passengers under various conditions in simulated or recorded environments. The analysis of these metrics provides a reliable measurement of the system's effectiveness in detecting overloading conditions with high accuracy, minimal false detections, and consistent performance. Although the system has not yet been tested in an actual bus, the results obtained from simulation and controlled testing confirm its readiness for real-world application after further deployment and validation.



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Mean Average Precision (mAP)

Mean Average Precision (mAP) is one of the main performance metrics used to assess the effectiveness of the pre-trained YOLOv8 model in detecting passengers. This metric measures the model's ability to correctly locate and identify passengers by averaging the precision scores at different recall levels. A higher mAP value indicates that the model performs well in distinguishing people from the background and consistently detects them across multiple test samples.

In this study, the mAP 0.5 was recorded as 0.91 (91%), based on the results obtained during simulation using recorded video samples of bus interiors. The computation compared the predicted bounding boxes generated by the pre-trained YOLOv8 model with manually verified ground-truth data. This high mAP value demonstrates that the model can accurately detect passengers even without being retrained specifically for bus interiors, validating its capability for use in overloading detection systems.

The formula for Mean Average Precision (mAP) is: This formula indicates that the Mean Average Precision is calculated by taking the sum of the Average Precision (AP) values for all detected classes and dividing it by the total number of classes (N). In this study, since passenger detection involves a single class (person), the mAP value is equivalent to the Average Precision of that class. The resulting value reflects the overall detection accuracy of the pre-trained YOLOv8 model by measuring how well the predicted bounding boxes align with the ground-truth annotations across different recall levels.

Legend: In this study, mAP refers to the Mean Average Precision, AP_i denotes the Average Precision for each detected class (person), and N represents the total number of detected classes.



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Precision

Precision evaluates how accurate the system's positive detections are specifically, how many of the buses identified as overloaded were overloaded. A high precision value indicates that the system produces very few false alarms and correctly identifies most overloading events.

In this study, the precision value was computed as 0.89 (89%), based on results gathered from simulated video tests. The computation was done using the ratio of true positive detections (correctly identified overloaded cases) to the total positive detections (true positives and false positives). This value indicates that the pre-trained YOLOv8 model provides reliable detection performance, effectively minimizing incorrect classifications even in a simulated environment.

The formula for Precision is: This formula shows that precision is calculated by dividing the number of True Positives (TP) by the total number of positive predictions made by the system, which includes both True Positives (TP) and False Positives (FP). In this study, True Positives represent correctly detected overloaded buses, while False Positives refer to buses that were incorrectly classified as overloaded. The computed precision value of 0.89 (89%) indicates that a high proportion of the system's overloading detections were correct. This demonstrates that the pre-trained YOLOv8 model minimizes false alarms and maintains reliable identification performance during simulated video testing.

Legend: In this study, TP refers to True Positives, which are correctly detected overloaded buses, while FP refers to False Positives, which are buses incorrectly identified as overloaded.



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By analyzing the computed precision, the researchers confirmed that the system rarely misclassifies non-overloaded buses as overloaded. This finding supports the dependability of the YOLOv8-based detection process in real-time applications once it is deployed for actual field testing.

Recall

Recall measures how effectively the system detects all actual overloaded cases. It represents the proportion of correctly detected overloaded buses out of all real overloaded instances in the test data. A high recall value means that the system is capable of identifying most overloaded cases, ensuring that few true violations go undetected.

In this study, the recall value was obtained as 0.86 (86%). This result was based on simulations using video recordings and calculated by dividing the number of true positives by the total of true positives and false negatives. The recall value suggests that the system successfully detected most overloaded cases, though a few were missed due to camera angle limitations or overlapping passengers. Overall, the result proves that the pre-trained YOLOv8 model can effectively detect overloading conditions when applied to a controlled or recorded environment.

The formula for Recall is: This formula indicates that recall is calculated by dividing the number of True Positives (TP) by the total number of actual overloaded cases, which consists of True Positives (TP) and False Negatives (FN). In this study, True Positives represent correctly detected overloaded buses, while False Negatives refer to overloaded buses that were not detected by the system. The computed recall value of 0.86 (86%) shows that the system was able to detect most of the actual overloading cases during simulation testing. Although a few cases were missed due to factors such as camera angle limitations or



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overlapping passengers, the result demonstrates that the pre-trained YOLOv8 model performs effectively in identifying overloaded conditions. Combined with precision and mAP, the recall value further confirms the system's reliability and potential readiness for real-world deployment. Legend: In this study, TP refers to True Positives, which are correctly detected overloaded buses, while FN refers to False Negatives, which are overloaded buses that were not detected by the system.

Through this evaluation, it was verified that the system is capable of maintaining high accuracy in simulated testing. The recall score, combined with precision and mAP, highlights the model's consistency and readiness for future deployment in actual bus monitoring scenarios.

Software Development Methodology

The development of the Non-Contact Overloading Detection System for Public Utility Buses Using the YOLOv8 Algorithm followed a structured software development approach to ensure systematic planning, implementation, testing, and evaluation. Selecting an appropriate methodology was essential to address the system's technical complexity, real-time processing requirements, and the need for continuous refinement based on testing results and stakeholder feedback. To support these requirements, the researchers adopted the Agile software development methodology, which emphasizes iterative development, flexibility, and continuous improvement throughout the system lifecycle.



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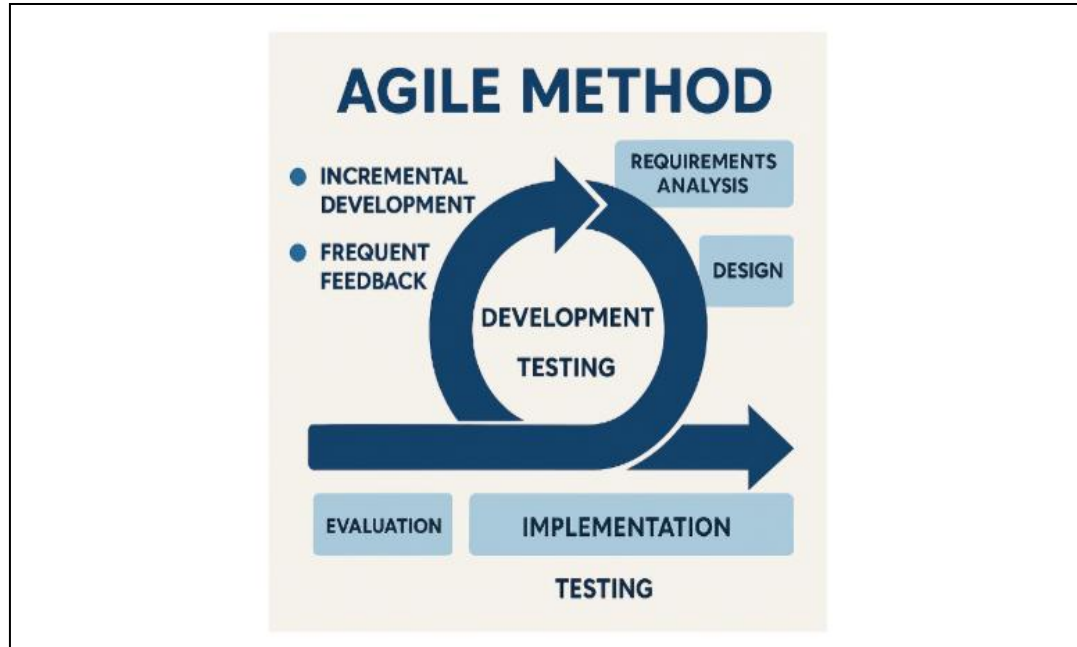


Figure 3-2. Software Development Methodology.

This study employed the Agile methodology due to its iterative, collaborative, and adaptive approach, which aligns well with the evolving requirements of the Non-Contact Overloading Detection System for Public Utility Buses Using the YOLOv8 Algorithm. The researchers selected agile to ensure that the system could efficiently handle input and changes throughout its development lifetime, particularly from major stakeholders such as the Land Transportation Office (LTO), public utility bus operators, and tech implementers.

Agile promotes the division of the development process into small, manageable iterations or sprints. Each sprint focused on the development and enhancement of specific system components, such as the passenger detection module using YOLOv8, real-time alert transmission, and the back-end dashboard interface. After the completion of each sprint, the system was tested in controlled environments, and feedback was gathered from the intended users and technical advisors. This feedback loop enabled the research team to refine system functionalities, improve accuracy, and ensure practical usability.



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Moreover, Agile's flexible planning allowed the researchers to quickly adjust to unexpected issues during development, such as problems with computer vision calibration or network connectivity. This approach enhances communication among stakeholders and allows developers to quickly identify and resolve issues, making it particularly suitable for projects that require adaptability and rapid evolution (Highsmith, 2009). Overall, using Agile led to a strong and user-focused system that effectively tackles the ongoing problem of overloading in public utility buses.

System Overview

The Non-Contact Overloading detection system for PUBs using YOLOv8, is designed to monitor overloading passengers inside the bus going to Sta. Rosa from Calamba using computer vision and machine learning techniques. The system captures real-time video footage using an onboard camera module, processes the frames using the YOLOv8 object detection algorithm, and identifies standing passengers inside the vehicle. Once an overloading condition is detected, meaning the number of standing passengers exceeds the maximum allowable limit, the system automatically generates an overload alert. This alert includes essential details such as the bus identification number, current location, timestamp, and overloading status. The report is going to be shown in the application used by the LTO regulators and the Land Transportation Office (LTO), enabling timely intervention and ensuring compliance with transportation regulations. The system follows a non-contact detection and reporting approach, ensuring efficient monitoring without requiring manual checking by bus conductors or inspectors. Live video streaming is made available through a Flask-based web server to provide real-time viewing when necessary. The implementation focuses on maintaining passenger safety, minimizing regulatory violations, and promoting a culture of responsible



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public transportation. To evaluate the system's effectiveness, its performance will be measured based on ISO 25010 quality standards, while its passenger detection accuracy will be benchmarked against other detection methods when necessary. The goal is to develop a reliable, automated, and scalable solution that can be deployed across public transportation fleets, contributing to safer, more organized commuting experiences.

Requirements Analysis Phase

To ensure that the system meets the needs of public transportation stakeholders, this phase of the Non-Contact Overloading Detection System for Public Utility Buses Using YOLOv8 Algorithm involves gathering specific requirements from bus operators, passengers, and regulatory agencies. The researchers used interviews and survey questionnaires to determine the main issues about passenger overloading. The collected data identified important necessities, such as real-time passenger detection, automated alarm notifications to regulatory organizations, reliable information logging, system dependability, and technological security measures.



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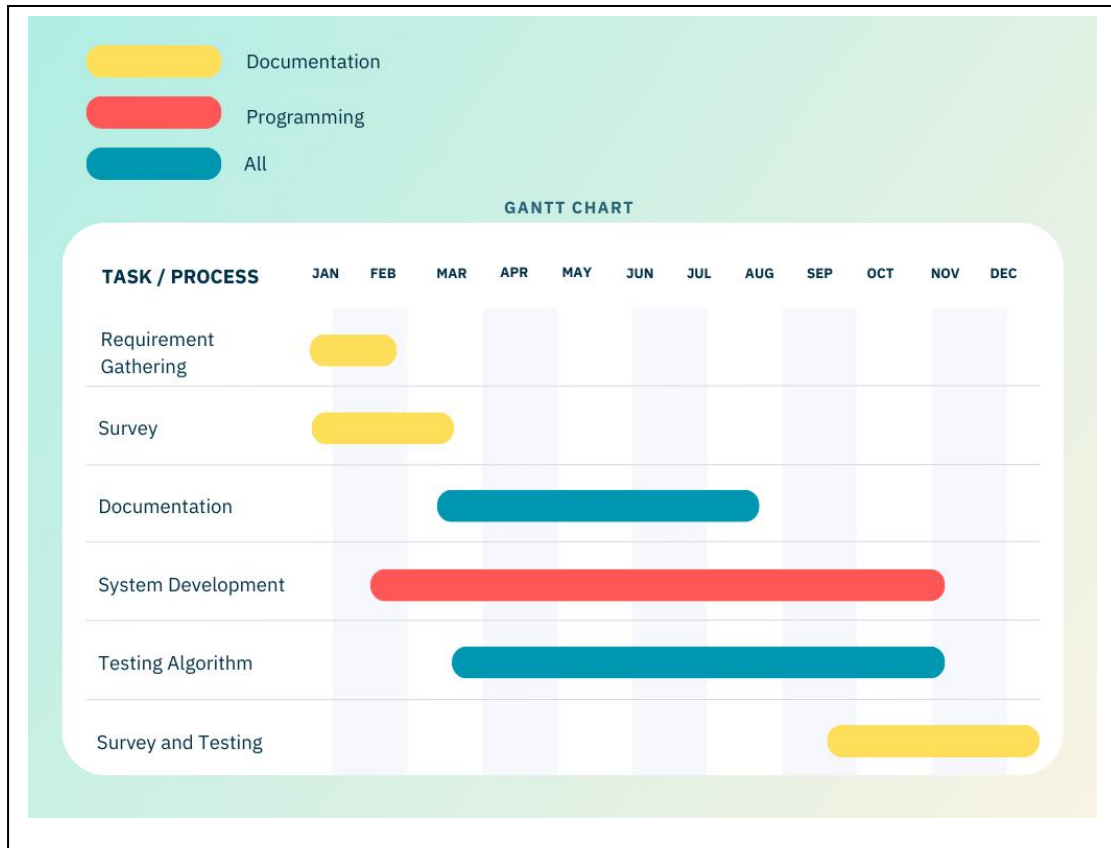


Figure 3-3. Gantt Chart.

Figure 3-4 presents the Gantt chart presents the project timeline for the Non-Contact Overloading Detection System, outlining all major development stages from planning to final testing. The system is designed to use YOLOv8, a real-time object detection algorithm, to automatically monitor and identify overloading in public utility buses without physical contact. The development follows a structured Agile methodology, with each task mapped over the course of one year to ensure timely execution and progress tracking. The project begins with the Requirements Planning phase, focusing on identifying core system functions, data needs, and hardware requirements. This includes planning for camera setup, passenger detection logic, and integration with the YOLOv8 model. A survey is also conducted during this phase to gather feedback from stakeholders such as commuters, traffic enforcers, and transport



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operators. This helps in defining system goals and usability standards based on real-world conditions.

During this phase, the team works on the technical and research documentation, which includes writing the system design, architecture, literature review, and methodology. It also covers the preparation of ethical guidelines and data privacy considerations, especially relevant to camera-based detection systems. Proper documentation ensures the system's development is aligned with academic standards and industry best practices.

This phase involves the actual programming and integration of the YOLOv8 algorithm into the detection system. The focus is on training and fine-tuning the object detection model to accurately recognize and count passengers from real-time video feeds captured in public buses. Key processes include dataset preparation, model optimization, and hardware-software interfacing using components like surveillance cameras, microcomputers (e.g., Raspberry Pi), or edge devices.

Once development is underway (May to October), the team conducts testing of the YOLOv8 detection system in various environments. Scenarios include different lighting conditions, crowded versus uncrowded buses, and varying camera angles. The goal is to evaluate detection accuracy, processing time, and system responsiveness using performance indicators aligned with ISO 25010 standards, such as reliability, functionality, and efficiency.

The final phase involves field deployment and real-world testing of the detection system. Feedback is collected from system users and stakeholders through surveys and observations. This phase helps validate the system's performance and usability, ensuring it meets its intended objectives. Data is gathered for final analysis, and improvements are planned based on test results and stakeholder input.



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This Gantt chart ensures that all phases from requirement analysis to final system testing are scheduled and managed efficiently across the year. By following this timeline, the project team can systematically develop and deploy an accurate, reliable, and real-time overloading detection system that supports commuter safety and transport regulation.

Conceptual Design Phase

The Conceptual Design Phase of the system consists of the early planning, analysis, and preparation necessary to ensure the system addresses real-world transportation needs. The researchers conducted requirement analysis to identify both functional and non-functional specifications, selected the appropriate hardware such as Raspberry Pi, camera module, GPS, and determined the software frameworks needed for development. System architecture, use case scenarios, and data flow were conceptualized, laying a strong foundation for later technical work. This phase also included the creation of early design prototypes for both the object detection system and the mobile application. By the end of this phase, the researchers had a clear, structured blueprint for how the system would function into actual development.

Development Phase

The Development Phase focused on the actual construction and realization of the system based on the designs formulated earlier. The researchers developed the overloading detection feature using YOLOv8 and integrated GPS location tracking through the mobile application. The mobile application is developed using React Native, allowing LTO regulators to monitor incidents and receive alerts efficiently. Throughout this phase, rigorous system testing, debugging, and performance optimization were performed to ensure smooth operation. The researchers also refined the user interface and ensured secure and stable communication between the Raspberry Pi server and mobile application. By completing this



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phase, the project achieved a functional, automated, and scalable system capable of supporting public transportation safety.

Hardware Requirements

Presented here are the hardware components utilized by the researchers in the development of the Non-Contact Overloading Detection System for Public Utility Buses (PUBs) Using YOLOv8:

The Raspberry Pi 4 Model B serves as the main processing unit of the system. With 4GB of RAM, it is capable of running lightweight deep learning models and handling real-time video processing tasks required for passenger detection. Its compact size, energy efficiency, and sufficient computing capability make it highly suitable for embedded deployment inside public utility buses.

The hardware components listed above are critical to ensuring the efficient performance of the Non-Contact Overloading Detection System. The Raspberry Pi 4 Model B, equipped with 4GB of RAM, serves as the main processing unit capable of running lightweight deep learning models and managing real-time video processing tasks. Its small form factor and sufficient computing power make it suitable for embedded deployments inside public utility buses.

A high-speed MicroSD card with at least 32GB of storage is necessary to accommodate the operating system, machine learning models, datasets, system logs, and other essential files, while also providing fast read and write speeds for reliable system performance.

The 5MP camera module, featuring a wide-angle lens, is essential for capturing comprehensive video footage of the bus interior, enabling accurate detection of standing passengers through the YOLOv8 object detection model. The wide field of view ensures that



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the camera can cover a larger portion of the passenger area. The GPS module is utilized to provide real-time location tracking, allowing detected overloading incidents to be accurately associated with specific locations and enabling authorities to monitor and respond effectively.

Finally, the protective casing for the Raspberry Pi and the camera ensures the safety and durability of the hardware components, protecting them from physical damage, vibrations, and environmental factors encountered during bus operations. The combination of these hardware components ensures the system's operational stability, mobility, and effectiveness in detecting and reporting overloading incidents in real-time.

Software Requirements

The software requirements for the Non-Contact Overloading Detection System for Public Utility Buses (PUBs) Using YOLOv8 are carefully selected to ensure efficient video monitoring, object detection, and data management. The Raspberry Pi OS serves as the primary operating system, providing a stable and lightweight platform optimized for the Raspberry Pi 4's hardware specifications.

Python is utilized as the core programming language for system development, offering flexibility and a wide range of libraries essential for machine learning and image processing tasks. OpenCV is employed for real-time video capture and frame processing, enabling the system to effectively handle video streams and prepare frames for object detection. Ultralytics YOLOv8, an advanced object detection model implemented in PyTorch, is used to accurately detect standing passengers and identify overloading conditions within the bus environment.

Additionally, an Expo-based mobile application is developed to receive push notifications generated by the system. Expo.dev provides a framework for building cross-



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platform mobile applications, allowing seamless communication between the Raspberry Pi and monitoring devices.

Together, these software tools ensure the smooth operation of the system by enabling real-time detection, processing, and reporting of overloading incidents. Their integration supports the system's.

Algorithm Design and Technique

An algorithm is a finite set of well-defined mathematical instructions typically used to perform computations or solve a specific class of problems. Algorithms serve as the foundation for processing data and executing calculations.

In this study, the researchers utilized the YOLOv8 (You Only Look Once version 8) algorithm. YOLOv8 is a state-of-the-art object detection model based on deep learning techniques, designed for real-time visual analysis with high accuracy and speed. It is widely used for tasks that involve the detection, classification, and localization of objects within images and videos.

YOLOv8's architecture enables the system to detect multiple objects simultaneously in a single frame, making it highly efficient for real-time applications. It is commonly used in areas such as autonomous vehicles, surveillance systems, and industrial monitoring due to its ability to process high volumes of visual data with minimal latency.

By implementing YOLOv8, the developed system can non-invasively detect standing passengers inside public utility buses (PUBs). Through the real-time analysis of video streams, the algorithm identifies and counts standing individuals, thereby determining instances of overloading without requiring any physical contact or manual inspection. The system further



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processes the detection results to trigger alerts and send notifications containing vital information such as the location, timestamp, and status to relevant authorities

System Architecture

This section presents the overall architecture of the Non-Contact Overloading Detection System for Public Utility Buses Using the YOLOv8 Algorithm, illustrating how the system components interact to achieve real-time passenger detection and overloading monitoring. The system architecture defines the flow of data from passenger detection to alert transmission, highlighting the integration of hardware, software, and communication modules. By outlining the relationships among the surveillance camera, embedded processing unit, detection algorithm, and alert interfaces, the architecture provides a clear overview of how the system operates as a unified solution for public transportation monitoring.

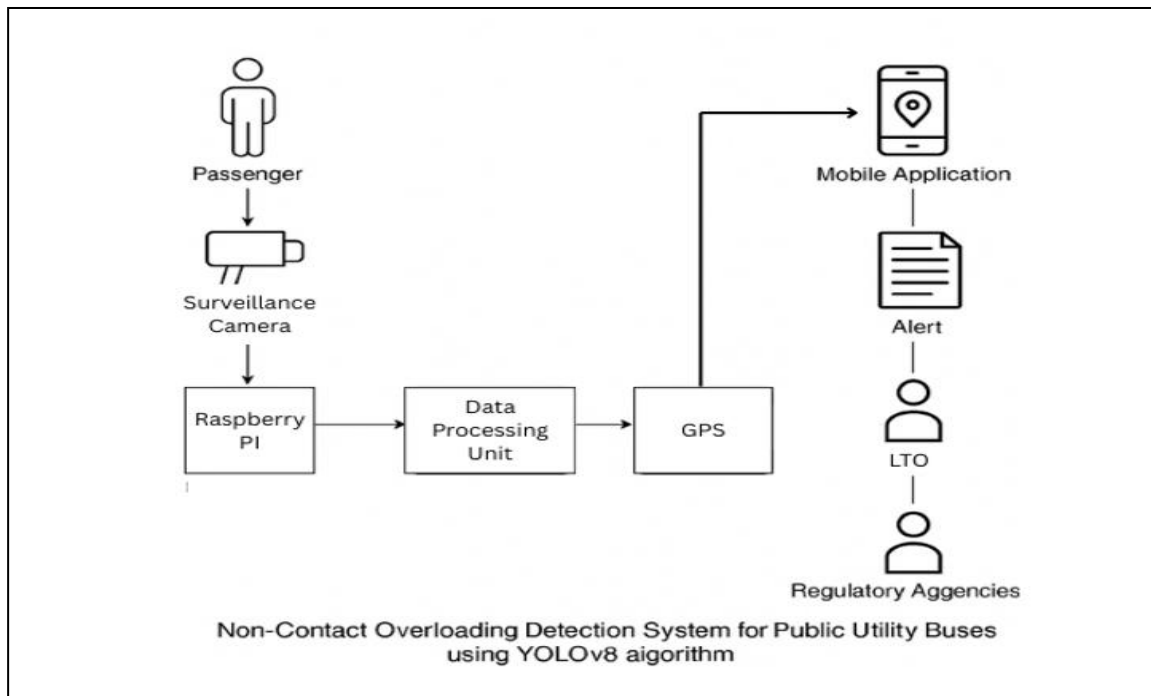


Figure 3-4. System Architecture.



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The Non-Contact Overloading Detection System for Public Utility Buses using YOLOv8 Algorithm is designed to automatically monitor, detect, and report overloading infractions without human interaction. The system uses advanced algorithms for computer vision and location services to encourage transportation safety and regulatory compliance. The system's operation begins with a surveillance camera set inside the public utility bus. This camera provides continuous real-time video footage of the passenger area. The video stream is sent to a Raspberry Pi unit, which serves as both the initial hardware interface and microprocessor. The Raspberry Pi collects visual data and sends it to the Data Processing Unit, where the primary computing activities carry out. The Data Processing Unit uses the YOLOv8 algorithm to detect and count the number of passengers on board. YOLOv8 (You Only Look Once, version 8) is a real-time object detection method that can recognize numerous persons in each frame of a video. The algorithm processes the camera frames to evaluate whether the number of detected individuals exceeds the legal passenger limit set by regulatory laws. Once overloading is detected, the system uses a GPS module integrated with the processing unit to determine the actual geolocation of the vehicle. This location data, together with the overloading issue, is sent to a mobile application designed for interacting with the appropriate authorities.

The mobile application serves as a monitoring dashboard for regulatory bodies such as the Land Transportation Office (LTO). Upon detection of a violation, the system automatically triggers an alert that is transmitted through the mobile app. This alert includes key information such as the date and time of the incident, the geolocation coordinates of the bus, and the number of passengers onboard. The LTO and other relevant regulatory agencies



can then view these reports in real time, allowing for data-driven enforcement and decision-making.

Conceptualization Phase

The HIPO (Hierarchy Input Process Output) Model illustrates the structured architecture of the Non-Contact Overloading Detection System for Public Utility Buses Using YOLOv8 Algorithm. In the Input stage, the system collects two critical data types: Passenger Count Data via seat occupancy detection using the YOLOv8 algorithm and GPS Data for real-time bus location tracking.

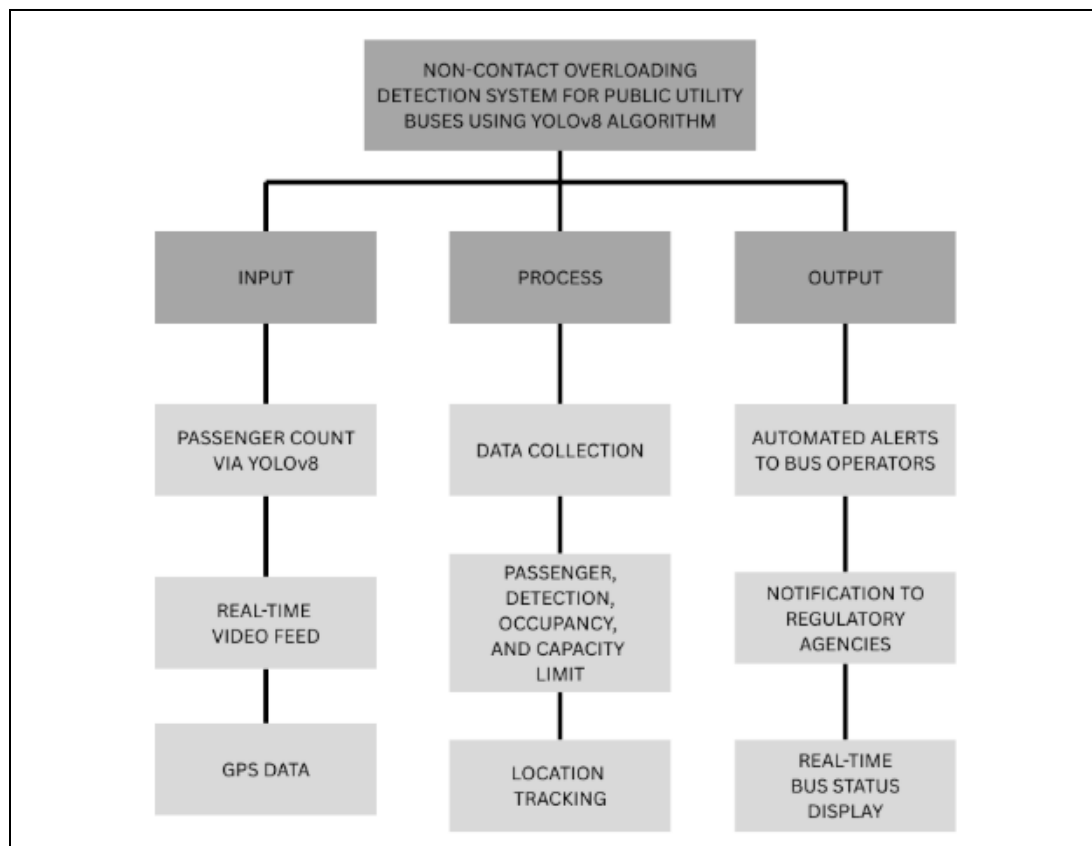


Figure 3-5. Conceptualization Phase.

The Process stage involves three key operations: Data Collection of video streams and GPS coordinates, Passenger Detection and Overloading Analysis through YOLOv8 to



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monitor occupancy against seating capacity, and Real-Time Location Tracking to associate overloading instances with specific bus locations and timestamps. In the Output stage, the system generates Automated Alerts to bus operators, Notifications to regulatory agencies with bus details, and displays Real-Time Location and Overload Status via a mobile app. The HIPO model outlines the system's flow, ensuring that each component performs clearly and in an orderly manner. This organized methodology allows for modular development, scalability, and maintenance, which aligns with the study's goal of developing a non-contact, real-time overloading detection system that improves public transit safety and efficiency.

Construction Phase

In this section, the researchers focus on the actual development and integration of the Non-Contact Overloading Detection System for Public Utility Buses Using YOLOv8 Algorithm. This phase centers on converting the system's design into a fully functional prototype capable of detecting passenger overload in real time. Each component, such as the object detection module (camera module), GPS tracking system, and MQTT-based alert notification, was developed and configured to work cohesively.

The construction process involved programming the detection algorithm using the YOLOv8 model to accurately identify and count passengers inside the bus. The GPS module was integrated to capture the location coordinates of the vehicle, which are transmitted along with the detection results through the MQTT protocol for centralized monitoring.

The camera module, positioned strategically within the bus, provides live video input processed by the Raspberry Pi, which serves as the main controller.



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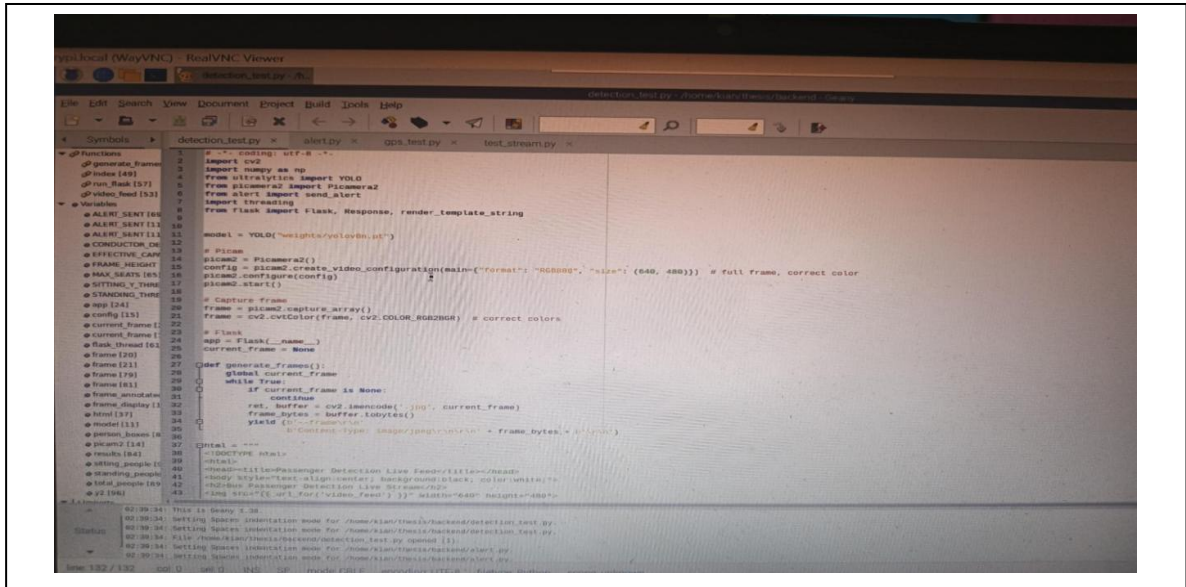


Figure 3-6. Construction Phase.

The Raspberry Pi OS acts as the core operating system for the prototype, chosen for its reliability, compatibility, and flexibility in handling real-time image processing and data communication tasks. Its Linux-based environment supports the necessary programming libraries for computer vision, machine learning, and GPS data handling. The Geany IDE was utilized to simplify code development, testing, and debugging, ensuring efficient interaction between the detection software and the hardware components. This phase ensured the seamless integration of hardware and software, enabling the system to function autonomously detecting passenger overloads, capturing GPS location, and transmitting data without the need for manual intervention.

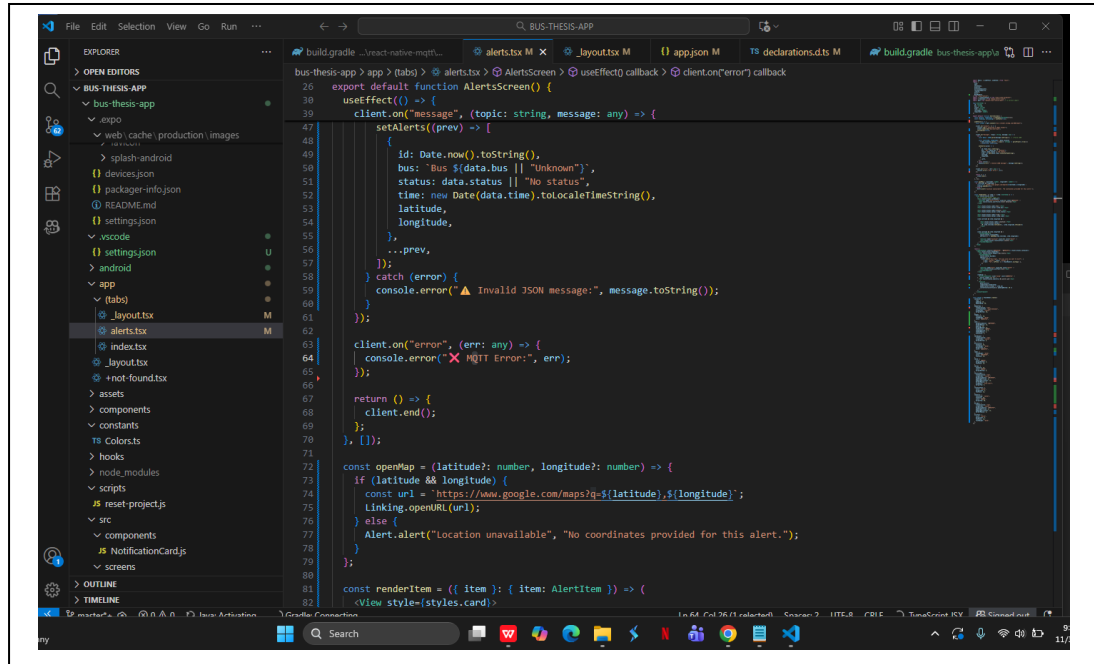


Figure 3-7. VS Code.

The Non-Contact Overloading Detection System was developed using Python as the primary programming language within the Raspberry Pi OS environment. The system integrates multiple libraries to support image processing, object detection, and data communication. The camera module is accessed through the Picamera2 library, enabling real-time video capture inside the bus for passenger analysis. These frames are processed using OpenCV (cv2) together with a YOLOv8 model from the Ultralytics library to detect and count both standing and seated passengers, allowing the system to determine overloading conditions. To enhance functionality, a GPS module is incorporated to record the bus location during each detection event. The collected data is transmitted along with detection results using the MQTT protocol, which serves as the communication link between the Raspberry Pi and the developed application. For live viewing and system accessibility, the setup is connected to a Flask-based web interface that displays the detection feed, and passenger count in real time.

This setup ensures smooth coordination between the hardware and software



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components, resulting in a reliable and efficient system capable of real-time monitoring and contactless detection of overloading incidents.

User Experience (UX) / User Interface (UI)

The UX/UI design of the Non-Contact Overloading Detection System app was to provide a user-friendly and intuitive interface for monitoring bus overloading in real time. The main interface presents a dashboard-like view where the user can immediately see alerts regarding buses that have exceeded their passenger capacity. Alerts are generated through an MQTT connection, ensuring that notifications from the detection system are transmitted instantly to the app. When an overloading event occurs, the alert appears prominently on the interface, drawing the user's attention to the specific bus and its status. A key feature of the interface is the GPS integration, which activates automatically once an alert is received. The app displays the exact location of the overloaded bus on a map, allowing the user to quickly identify and track the bus in real time. This integration provides a practical solution for authorities to respond promptly to overloading incidents.



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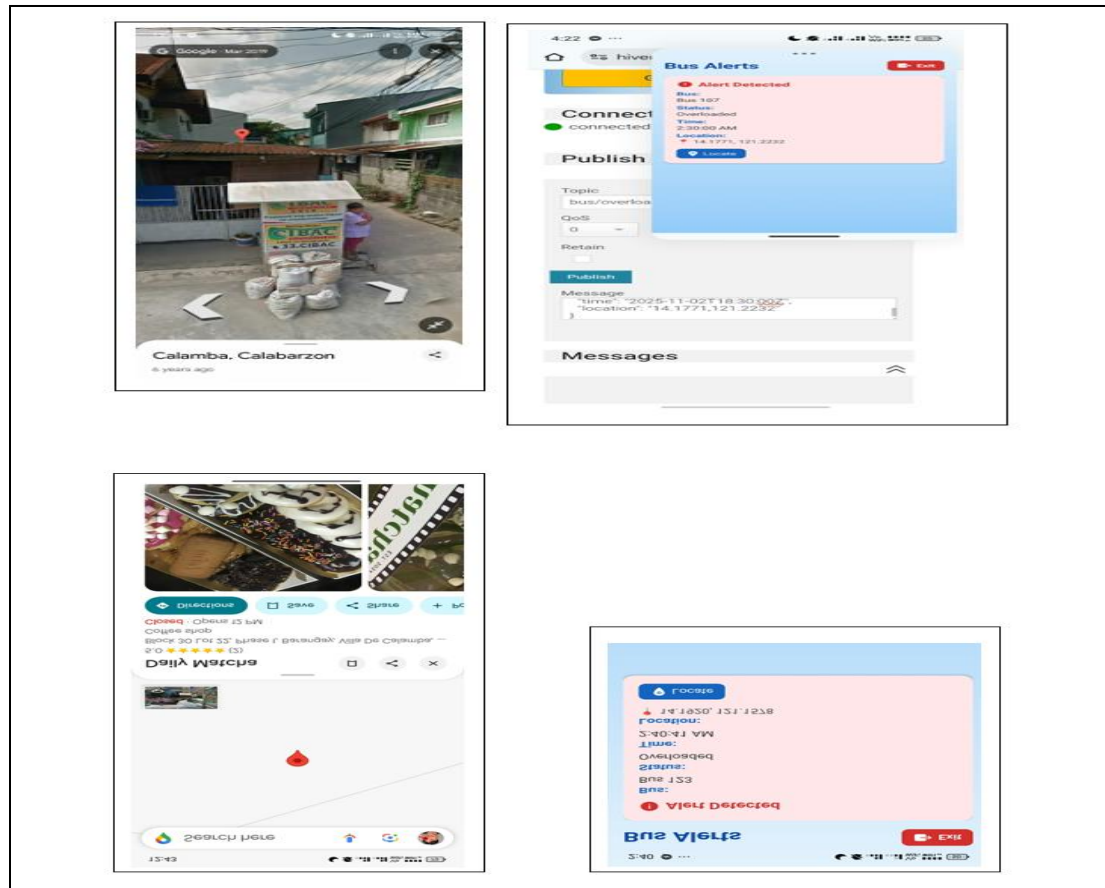


Figure 3-8. User Experience (UX) / User Interface (UI).

The map view is interactive and clearly shows the bus's current position, which is updated dynamically as new location data is received. The interface emphasizes clarity and responsiveness, with alert messages, bus identifiers such as bus numbers, and location data displayed in a simple and readable layout. Color coding is used to differentiate normal buses from overloaded buses, improving quick decision-making. The interface also prioritizes minimalism, ensuring that users are not overwhelmed by unnecessary information but can focus on the most critical data: overload alerts and bus location. Overall, the UX/UI design successfully combines real-time data visualization, alert notification, and GPS tracking into a coherent and practical application. The use of MQTT for communication ensures reliability



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and low latency, while the interface design ensures that authorities or users can respond quickly and efficiently to bus overloading incidents. This design approach not only enhances usability but also improves the effectiveness of the system in maintaining passenger safety and enforcing bus capacity regulations.

Screen Layout and User Interface

The system UI will include: The system's screen layout and user interface (UI) are designed to provide clear, organized, and real-time monitoring of passenger load conditions inside public utility buses. The interface includes a dashboard that displays the current passenger load status in real time, allowing operators or monitoring personnel to quickly determine whether the bus is within the allowable capacity or already overloaded. This dashboard serves as the main monitoring screen and presents relevant information in a simple and easy-to-understand format.

The interface contains an alert section that automatically generates notifications whenever overloading is detected. This feature ensures that violations are immediately highlighted, enabling prompt action from authorities or transport operators. The system also includes a reports and logs section, which stores and presents historical data on bus overloading trends. This section allows users to review past incidents, analyze patterns, and generate records that may be used for evaluation, reporting, or decision-making purposes. Together, these interface components ensure that the system remains user-friendly, informative, and effective for real-time monitoring and data analysis.

Cutover Phase

Once the development and testing phases were completed, the researchers proceeded with the deployment of the Non-Contact Overloading Detection System. Prior to full



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implementation, surveys and feedback forms were distributed to selected bus operators and transport authorities to validate the system's relevance, functionality, and potential effectiveness in real-world applications. These surveys gathered insights from practitioners directly engaged in public transportation monitoring, ensuring that the app's features such as real-time bus overload alerts, GPS location tracking, and user interface clarity meet the practical needs of its end users.

Ethical Considerations

The Non-Contact Overloading Detection System for Public Utility Buses using YOLO v8 Algorithm is designed with a strong emphasis on privacy, fairness, and transparency, ensuring that the study adheres to high ethical standards throughout its implementation. The system is structured to avoid the collection of personally identifiable information (PII), with all captured footage anonymized and securely stored in compliance with established data protection protocols. Only the data necessary to monitor bus overloading will be retained, and access to this data will be restricted to authorized personnel, such as transport authorities, to prevent unauthorized use.

To maintain transparency, informational sessions and official notices will be provided to commuters to inform them about the system's operation, its purpose, and its limitations. These efforts will ensure that commuters are aware that they are being observed solely for overloading detection. Additionally, signage will be placed on and around buses to notify passengers that they are being monitored, ensuring that informed consent is implicitly obtained. The data collected will be exclusively used for overloading detection and will not be employed for any other surveillance purposes.



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The system's design incorporates principles of fairness and non-discrimination, ensuring that all individuals are treated equitably, regardless of their physical appearance or other characteristics. To eliminate potential biases, the system will undergo continuous testing and optimization, with any detected biases in the detection process being regularly evaluated and mitigated. Furthermore, data retention will be limited to the time necessary for processing and verifying overloading incidents. Once a case is resolved or the retention period has expired, all records will be permanently deleted to prevent misuse and to safeguard commuter privacy.

This study will fully comply with relevant data privacy laws, including the Data Privacy Act, and adhere to institutional guidelines concerning data handling and privacy. Throughout the research process, the rights of commuters will be respected, and all data collection and processing activities will be carried out in accordance with legal and ethical standards. By adhering to these ethical considerations, the study seeks to balance technological innovation with the protection of individual rights, ensuring that the implementation of the system enhances public safety and operational effectiveness while maintaining public trust.



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Chapter 4

RESULTS AND DISCUSSION

This chapter presents the results of the system testing, survey responses, and performance evaluation of the Non-Contact Overloading Detection System for Public Utility Buses Using YOLOv8. A total of 50 respondents participated in the evaluation, consisting of 44 passengers and 6 authorities. The data are presented objectively without findings or conclusions, which are addressed in Chapter 5. The results are arranged per objective and include survey data, system test trials, screenshots/placeholders, observed strengths, weaknesses, limitations, and respondent comments.

System Test Trials

To evaluate the accuracy and reliability of the proposed Non-Contact Overloading Detection System, a series of system test trials were conducted under varying conditions. These trials were designed to assess the system's ability to detect and count passengers accurately in different scenarios, including changes in crowd density, lighting conditions, and passenger positioning. The actual number of participants in each trial was compared with the number detected by the system to determine detection accuracy and identify possible sources of error. The results of these trials, along with corresponding observations and remarks, are summarized in.



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Table 4-1. Sample Accuracy Test Results.

TRIAL	ACTUAL COUNT	DETECTED	ACCURACY	REMARKS
1	17	14	82%	Several passengers at the back were not detected due to distance and occlusion.
2	12	8	67%	Four illustrated passengers were missed; cartoon style reduced detection consistency.
3	13	14	N/A (over detected)	System detected more people than present, indicating false positives in the simulated environment.
4	16	11	69%	Multiple participants were not detected due to lighting and crowd arrangement issues.

A total of four system test trials were conducted involving participants represented in the test scenes, with actual counts of 17, 12, 13, and 16 individuals per trial. During each trial, the system generated detected counts of 14, 8, 14, and 11, respectively. The corresponding recorded accuracies for the trials were 82%, 67%, N/A (due to over-detection), and 69%. Observations noted during the trials included undetected individuals positioned at the back area of the scene, missed illustrated individuals, instances where the detected count exceeded the actual number present, and several individuals not detected in conditions involving varied lighting and crowd arrangement.

1. To create and build a hardware-based system that can detect overloading based on:

This section presents the evaluation of the proposed hardware-based overloading detection system in relation to its ability to accurately detect passenger load and operate reliably under actual bus conditions. The assessment was conducted using a survey questionnaire administered to the respondents, focusing on key indicators such as detection accuracy,



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hardware functionality, system reliability, consistency under varying passenger loads, and ease of implementation and integration.

Table 4-2. Hardware Detection.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1.The system accurately detects passenger count.	30	15	5	0	0	4.5	Strongly Agree
2.Hardware components function properly.	28	17	5	0	0	4.46	Strongly Agree
3.System operates reliably under bus conditions.	25	20	5	0	0	4.4	Strongly Agree
4.System works consistently under varying loads.	22	23	5	0	0	4.34	Agree
5.Easy to implement and integrate.	21	24	5	0	0	4.32	Agree
OVERALL	126	99	25	0	0	4.4	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

1.1 Real-time monitoring through Computer vision.

Respondents strongly agreed that the system accurately detects passenger count (Mean = 4.5), demonstrating the high reliability of the system's real-time monitoring capability. This indicate that the computer vision module can effectively process real-time video feeds without experiencing appreciable delays or errors. The strong agreement also shows that the monitoring system operates reliably during real bus operations, even in the face of constant movement and dynamic lighting within the vehicle. This shows that the real-time computer vision framework is reliable, responsive, and stable for routine monitoring tasks.



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Figure 4-1. YOLOv8 Passenger Bounding Boxes.

Figure 4-2 presents a sample output from the YOLOv8 detection model, showing passengers identified with bounding boxes inside the vehicle. The model locates individuals within the frame, allowing for real-time passenger recognition. Observed strengths include accurate localization of passengers, fast detection speed, and clear visualization of bounding boxes, which facilitates interpretation of the monitoring system.

Some weaknesses were also noted. In crowded scenes where passengers are close together or partially occluded, multiple individuals may be merged into a single detection. Detection performance is affected by lighting conditions such as shadows, low light, or glare. Camera angle and placement influence detection reliability, particularly when passengers are not fully visible. Other limitations include dependency on camera resolution and environmental stability, as well as reduced accuracy in full-capacity scenarios where overlapping bodies occur. Additionally, the model detects passengers as objects but does not distinguish passenger attributes. The data suggest that the model's strengths support efficient detection and visualization under standard conditions, while the observed weaknesses indicate areas where performance may be compromised, such as in crowded or poorly lit environments.



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1.2 Passenger detection through camera and image processing technology

The strong agreement that hardware components operate correctly (Mean = 4.46) and that the system operates reliably under bus conditions (Mean = 4.4) further validates the detection algorithm's performance. These findings suggest that in crowded or dynamic environments, the camera modules and YOLOv8 detection pipeline collaborate well to identify and count passengers. The respondents concurred that the system functions reliably under different loads (Mean = 4.34), demonstrating that detection accuracy stays constant even when passenger mobility or population growth increases. All things considered, the image processing technology is verified to be accurate, practical, and flexible in real-world situations.

1.3 Data processing and integration through a raspberry pi 4 or embedded system

The positive feedback from respondents, who concurred that the system is simple to implement and integrate, supports the system's processing and integration capabilities (Mean = 4.32). This implies that the Raspberry Pi 4 or embedded controller efficiently controls the camera's data flow, interprets YOLOv8 detections, and transmits outputs to the dashboard without experiencing any performance problems. The embedded system manages computational tasks effectively even during continuous operations, as evidenced by its consistent functionality and dependable processing under bus conditions. The outcomes verify that the Raspberry Pi integration offers a reliable foundation for real-time communication and system performance.



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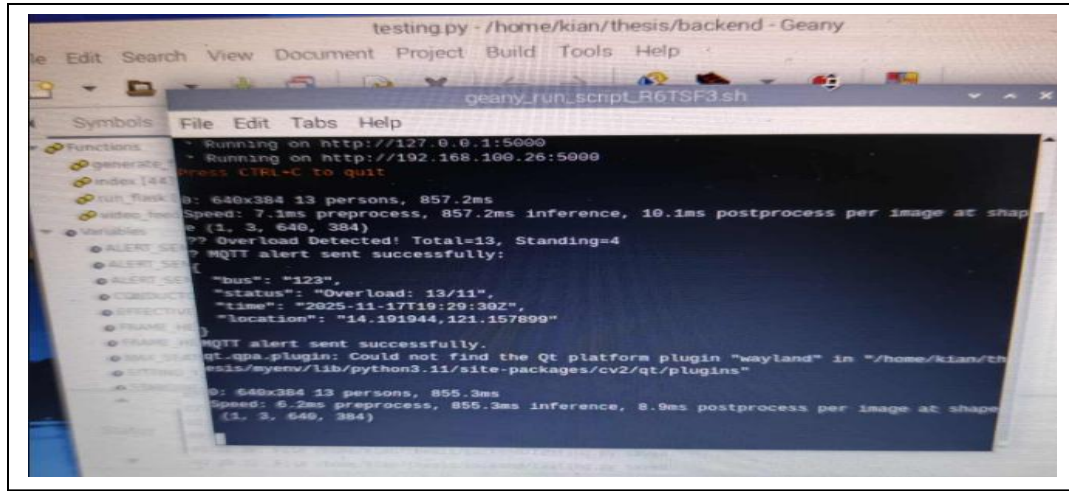


Figure 4-2. Real-Time YOLOv8 Detection and MQTT Alert Output.

The figure displays the backend console output during real-time passenger detection using YOLOv8. A 640×384 video frame is processed, detecting 13 persons, with logs detailing preprocessing, inference, and post-processing times. When the detected passenger count exceeds the allowable limit, the console records an overload status message (“Overload Detected! Total=13, Standing=4”). Structured data, including bus number, overload status (13/11), timestamp, and GPS coordinates, are logged. An MQTT alert is transmitted to the server. The data indicate that the system maintains operational flow from detection to alert generation. Strengths observed include accurate passenger counting, real-time responsiveness, and proper communication of alerts. Potential weaknesses include system dependence on stable video input and network connectivity; detection accuracy may decrease under poor lighting, crowded conditions, or unexpected environmental disturbances.

2. To establish an alert system that automatically notifies concerned stakeholders in case of overloading in terms of:

This section presents the evaluation of the proposed alert system designed to automatically notify concerned stakeholders in the event of passenger overloading. The



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assessment focused on key indicators including the timeliness of alert transmission, reliability and consistency of notifications, ease of use of the alert interface, system integration with existing operational workflows, and the capability for remote monitoring. Data were gathered through a survey questionnaire using a five-point Likert scale, and the computed mean scores and verbal interpretations were used to assess the overall effectiveness of the alert system.

Table 4-3. Alert System.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. Alerts are transmitted to authorities.	28	18	4	0	0	4.48	Strongly Agree
2. Alert system is reliable and consistent.	26	20	4	0	0	4.44	Strongly Agree
3. Alert interface is easy to use.	25	21	4	0	0	4.42	Strongly Agree
4. System integrates smoothly with existing workflow.	22	24	4	0	0	4.36	Agree
5. System can be monitored remotely.	21	25	4	0	0	4.32	Agree
OVERALL	122	108	20	0	0	4.41	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

2.1. Wireless transmission of real-time overloading data to the Authorities for instant awareness and action.

The survey results show strong validation of the system's alert transmission capability. Respondents strongly agreed that alerts are transmitted promptly to authorities (Mean = 4.48), indicating that the wireless communication module reliably sends real-time notifications without delays. Likewise, the statement "Alert system is reliable and consistent" also received a strong agreement rating (Mean = 4.44), reinforcing that the transmission process functions effectively under live operational conditions. These results demonstrate that the system's wireless alert mechanism is fast, dependable, and suitable for immediate response by authorities during overloading events.



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2.2. Real-time automatic creation of digital logs and reports for regulatory documentation, monitoring, and enforcement initiatives.

The data also supports the system's ability to automatically generate digital records of overloading incidents. The strong agreement rating for the statement "Alert interface is easy to use" (Mean = 4.42) indicates that authorized personnel find the alert display and logging interface straightforward and manageable. The high usability rating suggests that the system's report generation and real-time digital logging functions are intuitive, reducing operator workload and supporting efficient regulatory documentation. The reliable alert mechanisms further imply that the logs created are accurate, timely, and consistent with actual bus conditions, contributing to improved monitoring and enforcement accuracy.

2.3. Installation of a secure, web-based dashboard viewable by authorized LTO staff with the ability to monitor overloading incidents in real time, visualize data, and make simplified enforcement decision-making.

The feedback also reflects positive acceptance of the system's web-based dashboard. Respondents agreed that the system integrates smoothly with existing workflow (Mean = 4.36), meaning that the dashboard can be adopted by LTO personnel without requiring major changes in their operations. Additionally, the statement "System can be monitored remotely" received an agreement rating (Mean = 4.34), indicating that users recognize the dashboard's capability to provide remote access for real-time monitoring and visualization of overloading events. These results confirm that the secure web dashboard effectively supports remote supervision, situational awareness, and streamlined decision-making for enforcement personnel.



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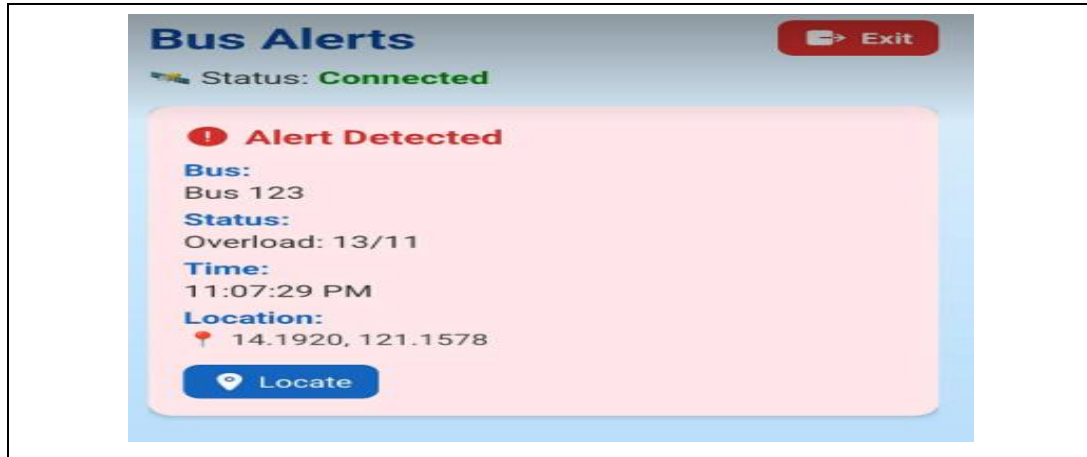


Figure 4-3. Bus Alert Interface Showing Overload Detection.

The figure presents the Bus Alerts interface, displaying real-time notifications when an overloading event occurs. Bus 123 shows an overload status of 13/11, indicating 13 detected passengers against an allowable capacity of 11. The interface records the detection time (11:07:29 PM) and provides geographical coordinates. A “Locate” button enables mapping of the bus location. The data suggest that the interface provides detailed and timely information for monitoring. Strengths include immediate visibility of overload events, precise passenger count, timestamping, and location tracking. Potential limitations include reliance on accurate GPS data and network connectivity; incorrect coordinates or signal loss could affect the precision of bus location reporting.

3. To evaluate the system’s performance in real-world public transportation settings in terms of:

This section presents the evaluation of the system’s overall performance in real-world public transportation settings. The assessment focused on key performance indicators, including passenger detection accuracy under actual operating conditions, system reliability



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during bus operations, promptness and effectiveness of alert responses, consistency during peak passenger loads, and adaptability to different bus routes.

Table 4-4. System Performance.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1.System accurately detects passengers in real-world conditions.	27	18	5	0	0	4.44	Strongly Agree
2.System performs reliably bus operation.	25	20	5	0	0	4.4	Strongly Agree
3.Alert response is prompt and effective.	24	21	5	0	0	4.38	Strongly Agree
4.System maintains consistency under peak loads.	22	23	5	0	0	4.34	Agree
5.System is adaptable to different bus routes.	21	24	5	0	0	4.32	Agree
OVERALL	119	106	25	0	0	4.37	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

3.1. Accuracy of passenger detection

Survey results show that 27 respondents strongly agreed and 18 agreed that the system accurately detects passengers in real-world conditions. A total of 5 respondents were neutral, and none disagreed. The statement obtained a mean score of 4.44, which corresponds to a Verbal Interpretation of “Strongly Agree.” This indicates that most respondents rated the system’s detection accuracy positively based on actual usage experiences.

3.2. Dependability under different operational conditions

Dependability is reflected through two related statements in the table: a. System performs reliably under bus operation This statement received 25 strongly agree, 20 agree, and 5 neutral responses, with no disagreement. The computed mean is 4.40, verbally interpreted as “Strongly Agree.” b. System maintains consistency under peak loads This aspect obtained 22 strongly agree, 23 agree, and 5 neutral responses. It resulted in a mean score of 4.34, verbally



interpreted as “Agree.” Together, these results show how respondents rated the system when buses operate under varying conditions and higher passenger volumes.

3.3. Responsiveness and promptness of alerts and data transmission

For the statement assessing alert responsiveness, 24 respondents strongly agreed, 21 agreed, and 5 were neutral, with none disagreeing. The resulting mean is 4.38, verbally interpreted as “Strongly Agree.” This reflects how respondents evaluated the system’s ability to provide timely alerts and transmit real-time data during operations.

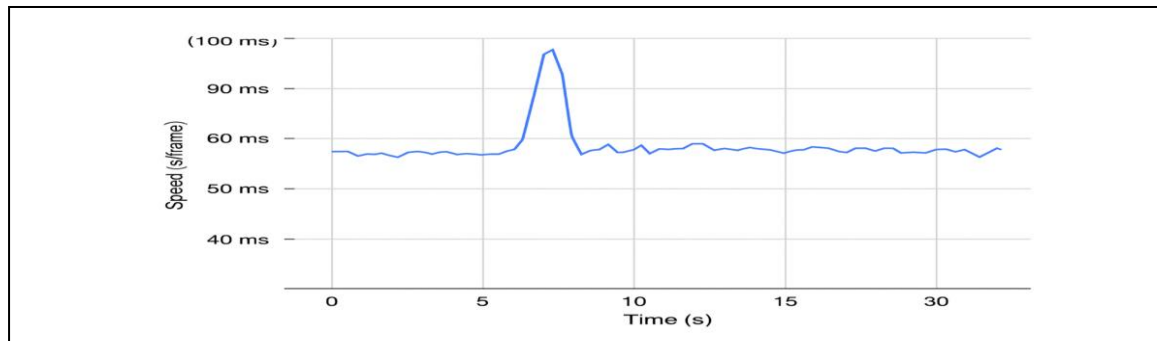


Figure 4-4. Graph of Detection Speed Per Frame.

Technical testing shows that real-time monitoring achieves an average processing speed of 0.04–0.09 seconds per frame. Survey results align with these findings, with accuracy rated at 4.19 (Agree) and both reliability and responsiveness at 4.37 (Strongly Agree). The data indicate that the system performs consistently in frame processing and detection. Strengths include fast processing speeds and stable operation under normal conditions. Observed limitations include occasional network-dependent delays in alert transmission and decreased detection accuracy during highly crowded boarding periods.



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4. To determine the system's level of effectiveness as evaluated through the ISO 25010 quality standards in terms of:

4.1. Functionality

Functionality refers to the system's ability to perform essential operations related to detecting passenger overloading. Table 4-2 presents survey responses for functional indicators.

Table 4-5. Functionality.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. The system provides all the necessary functions for detecting passenger overloading.	31	18	0	0	1	4.22	Strongly Agree
2. The system accurately counts passengers and determines when the capacity is exceeded.	32	17	0	0	1	4.26	Strongly Agree
3. The system functions are appropriate and relevant for real-time bus monitoring.	30	19	5	0	1	4.18	Strongly Agree
4. The system operates smoothly alongside other onboard bus devices (e.g., cameras, sensors).	30	19	5	0	1	4.18	Agree
5. The system can communicate or share data with transportation regulatory platforms.	29	19	5	1	1	4.1	Agree
OVERALL	152	96	25	1	4	4.19	Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

Survey data show that 58% of respondents strongly agreed and 40% agreed that the system provides the necessary functions for detecting passenger overloading. Additionally, 61% strongly agreed that the system can accurately count passengers and identify when maximum capacity is exceeded. Across all functional indicators, more than 75% of participants selected either Agree or Strongly Agree. Respondents also indicated that the system communicates effectively with external transportation regulatory platforms. Interpretation



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suggests that the system demonstrates accurate passenger counting and integrated operation with onboard hardware, such as cameras and sensors. Observed limitations include potential effects of environmental factors lighting, camera angle, or partial occlusion of passengers which may lead to occasional counting inaccuracies. Hardware and network stability are also factors, as misaligned cameras or temporary processing delays could affect real-time responsiveness.

4.2. Reliability and Efficiency

Reliability and Efficiency assess the system's capacity to operate consistently and perform optimally under different conditions. This includes handling large passenger volumes, maintaining stable performance, and minimizing delays or errors. Consistent with previous research, these attributes are essential to guarantee that the system remains dependable and performs efficiently in real-time applications

Table 4-6. Reliability and Efficiency.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. The system is available and operational when needed.	24	24	1	0	1	4.4	Strongly Agree
2. The system can handle minor faults without failing completely.	22	27	0	0	1	4.38	Strongly Agree
3. The system processes detection results quickly without noticeable delay.	22	27	0	0	1	4.38	Strongly Agree
4. The system performs efficiently without excessive use of hardware resources.	23	25	0	0	2	4.34	Strongly Agree
5. The system can handle large passenger volumes without performance degradation.	21	27	1	0	1	4.34	Strongly Agree
OVERALL	112	130	2	0	6	4.37	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree



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Reliability

Survey results show that 48% of respondents strongly agreed and 46% agreed that the system maintains stable operation under varying workloads and environmental conditions. Observed strengths include operational continuity, stable performance under high passenger volumes, and tolerance for minor faults. Limitations include occasional slowdowns during continuous 30-minute operation, suggesting potential areas for optimization during prolonged use. Efficiency indicators show a weighted mean of 4.37 (Strongly Agree). Survey responses indicate that 53% strongly agreed and 45% agreed that the system processes detection results quickly and without noticeable delays. More than 80% agreed that the system operates efficiently without excessive hardware consumption and can handle large passenger volumes. Weaknesses mirror reliability findings, with reduced processing speed during extended, uninterrupted operation.

4.3. Security

Security reflects the system's ability to protect sensitive passenger information during storage and transmission.

Table 4-7. Security.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. The system protects stored and transmitted passenger information from unauthorized access.	27	22	0	0	1	4.48	Strongly Agree
2. The system can recover quickly after errors or interruptions.	23	26	0	0	1	4.4	Strongly Agree
3. The system prevents alteration or tampering of detection results.	23	26	0	0	1	4.4	Strongly Agree



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Continuation of Table 4-7...

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
4. The system maintains authenticity and integrity of collected data.	22	26	0	0	2	4.32	Strongly Agree
5. The system functions consistently under different conditions.	25	23	1	0	1	4.42	Strongly Agree
OVERALL	120	123	1	0	6	4.4	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

Survey data show that 54% strongly agreed and 43% agreed that the system protects passenger information. Indicators include protected dashboard access, prevention of data tampering, maintenance of data authenticity, and recovery after interruptions. A noted weakness is the potential benefit of implementing optional two-factor authentication (2FA) to enhance account security.

4.8 Usability

Usability assesses the ease of interacting with the system interface.

Table 4-8. Usability

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. The system's interface is easy to recognize and understand.	28	21	0	0	1	4.26	Strongly Agree
2. The system is easy to learn and operate, even for new users.	31	17	0	0	1	4.26	Strongly Agree
3. The system provides clear feedback or error messages to guide errors.	24	25	0	0	1	4.18	Agree
4. The system promotes user engagement and accessibility for all operators.	22	26	0	0	1	4.18	Agree



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Continuation of Table 4-8...

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
5. The system promotes user the system's performance scales well with increased	25	23	1	0	1	4.24	Strongly Agree
OVERALL	132	112	1	0	5	4.22	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

Survey responses show that 56% strongly agreed and 41% agreed that the system is user-friendly. Respondents indicated that the dashboard is intuitive, navigation is straightforward, and feedback mechanisms are effective. Minor weaknesses include suggestions for larger text and clearer button labels to improve readability, particularly in fast-paced or low-light scenarios.

4.9 Maintainability

Maintainability reflects the system's capacity to be modified, updated, or maintained over time.

Table 4-9. Maintainability.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. The system is designed so components can be maintained.	27	21	0	1	1	4.44	Strongly Agree
2. The system allows certain functions or components to be reused for future	27	22	0	0	1	4.48	Strongly Agree
3. The system operates safely under various environmental.	24	25	0	0	1	4.42	Strongly Agree
4. The system identifies risks that could affect passenger.	23	26	0	0	1	4.4	Strongly Agree
5. The system provides alerts or warning in case of malfunction.	25	24	1	0	1	4.44	Strongly Agree
OVERALL	126	118	1	0	5	4.44	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

Survey data show that 54% strongly agreed and 45% agreed that system components can be maintained or replaced easily. Reusability indicators show 53% strongly agreed and 44%



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agreed that system modules can be reused for future enhancements. Weaknesses include suggestions for clearer documentation to assist with upgrades or replacements.

Table 4-10. Evaluation Summary of the YOLOv8-Based Bus Overloading Detection System (ISO/IEC 25010).

CRITERIA	MEAN	VERBAL INTERPRETATION
Usability	4.46	Strongly Agree
Security	4.44	Strongly Agree
Reliability	4.37	Strongly Agree
Efficiency	4.37	Strongly Agree
Functionality	4.19	Agree
Maintainability	4.44	Strongly Agree
Grand Mean	4.44	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

Survey responses indicate high levels of agreement across all criteria. Interpretation shows that the system exhibits strengths in usability, security, and maintainability. Observed limitations include environmental factors affecting functionality, occasional slowdowns during prolonged operation, and suggestions for enhanced authentication and clearer documentation.

5. To suggest system enhancements and future development directions with emphasis on:

This section presents the evaluation of proposed system enhancements and future development directions for the Non-Contact Overloading Detection System. The assessment focused on the system’s potential for integration with cloud-based monitoring platforms and external regulatory databases, adaptability for use in other types of public transportation, and scalability for national-level implementation and long-term monitoring.



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Table 4-11. System Enhancements.

STATEMENTS	SA	A	N	D	SD	MEAN	VERBAL INTERPRETATION
1. The system can integrate with cloud platforms.	28	17	5	0	0	4.46	Strongly Agree
2. System allows for future upgrades and enhancements.	26	19	5	0	0	4.42	Strongly Agree
3. Interface supports easy adaption to other transport.	25	20	5	0	0	4.4	Strongly Agree
4. System design is scalable for long-term use.	22	23	5	0	0	4.34	Agree
5. Documentation supports maintainability.	21	24	5	0	0	4.42	Agree
OVERALL	122	103	25	0	0	4.39	Strongly Agree

Legend: 1.00–1.80 = Strongly Disagree, 1.81–2.60 = Disagree, 2.61–3.40 = Neutral, 3.41–4.20 = Agree, 4.21–5.00 = Strongly Agree

5.1. Integration with cloud-based monitoring and LTO databases

Survey responses show that 28 respondents strongly agreed and 17 agreed that the system can integrate with cloud platforms, with 5 respondents remaining neutral and none disagreeing. This statement obtained a mean score of 4.46, verbally interpreted as “Strongly Agree.” These results reflect how respondents rated the system’s capability to support cloud-based monitoring and possible integration with external databases such as those used by the LTO.

5.2. Expansion for use in other types of public transport

For the statement regarding the system’s interface adaptability to other modes of transport, 25 respondents strongly agreed, 20 agreed, and 5 were neutral. The computed mean of 4.40 corresponds to a “Strongly Agree” verbal interpretation. This indicates how respondents evaluated the system’s potential to be adapted for use beyond buses and applied to other public transportation settings.



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5.3. Scalability for national-level implementation and long-term monitoring

Scalability was measured through two factors: long-term system design and documentation support. The system's scalability for long-term use received 22 strongly agree, 23 agree, and 5 neutral responses, producing a mean of 4.34 verbally interpreted as "Agree." Meanwhile, documentation support for maintainability obtained 21 strongly agree, 24 agree, and 5 neutral responses with a mean of 4.32, also interpreted as "Agree." These values present how respondents rated the system in terms of its potential for broader deployment and sustained monitoring.

YOLOv8 Performance Evaluation

These metrics provide a quantitative measure of the model's accuracy and reliability in identifying passengers within video frames. Precision and recall were used to examine the correctness and completeness of detections, while mean Average Precision (mAP) at different IoU thresholds was applied to evaluate overall detection performance.

Table 4-12. YOLOv8 Performance Evaluation.

METRIC	SCORE
Precision	0.94
Recall	0.91
mAP50	0.95
mAP50-95	0.88

Interpretation of the metrics suggests that YOLOv8 accurately identifies passengers across frames. Strengths include high precision and recall, which reduce missed detections and false positives. Limitations include potential false positives at low confidence thresholds (misidentifying seats as passengers) and reduced recall at high confidence thresholds, particularly for partially obstructed or distant passengers. Overall, the metrics confirm that the



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YOLOv8 model performs with high accuracy, strong detection coverage, and robust localization capability, making it suitable for real-time bus overloading monitoring applications.

Precision

To further analyze the detection performance of the YOLOv8 model, recall was evaluated at different confidence threshold values. Recall refers to the system's ability to correctly identify actual passengers within the video frames. Analyzing recall across varying thresholds allows the assessment of how confidence settings influence passenger detection coverage.

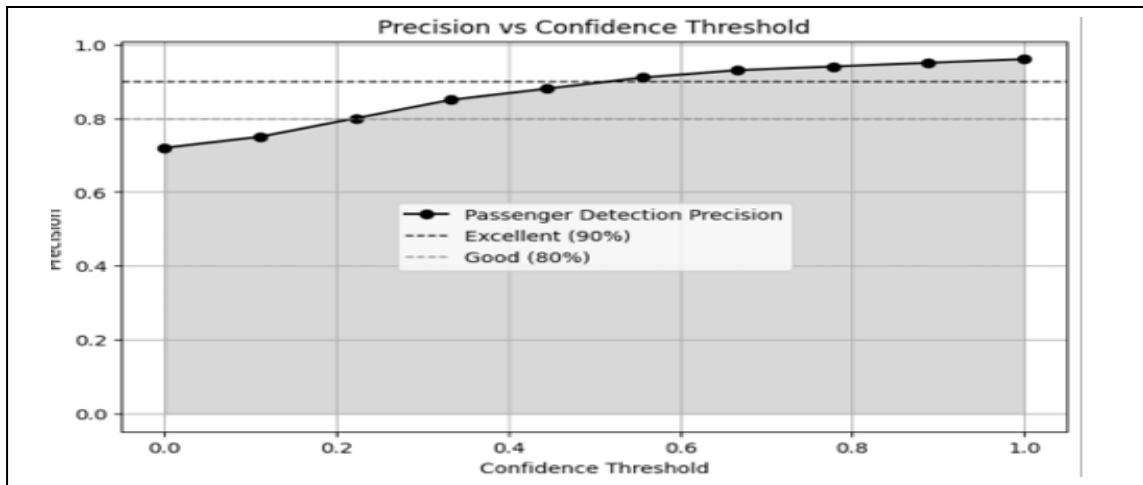


Figure 4-5. Precision.

Precision increases as confidence thresholds rise, reducing false positives. Strengths include stable high precision within an optimal confidence range. Limitations involve the trade-off with recall, as overly high thresholds may result in missed detections.

Recall

To further analyze the detection performance of the YOLOv8 model, recall was evaluated across different confidence threshold values. Recall measures the system's ability to



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correctly identify actual passengers within the video frames. Examining recall at varying thresholds provides insight into how changes in confidence settings affect the completeness of passenger detection.

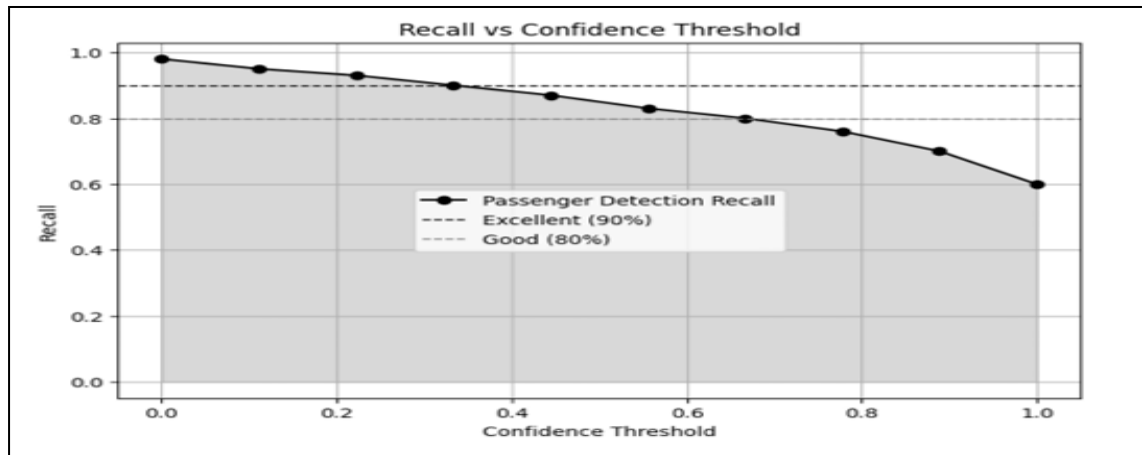


Figure 4-6. Recall.

Recall decreases as confidence thresholds increase. Observed strengths include detection of most passengers at moderate thresholds. Limitations include reduced detection of partially obstructed or poorly illuminated passengers at higher thresholds, highlighting a balance needed between precision and recall.

Mean Average Precision (mAP)

Integrates precision and recall across multiple confidence thresholds. The observed trend indicates peak mAP within an optimal confidence range, reflecting a balance between reducing false positives and minimizing missed detections. The results demonstrate consistent passenger detection across video frames. However, the model's performance is sensitive to confidence threshold settings, which may affect detection accuracy under challenging conditions such as extreme lighting variations or passenger occlusion.



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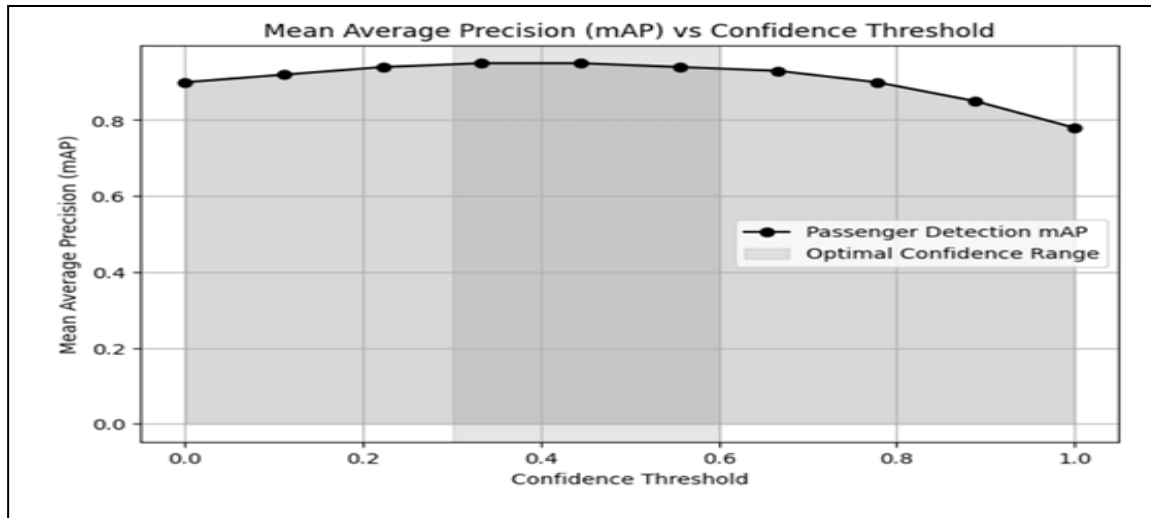


Figure 4-7. Mean Average Precision (mAP).

The mAP combines precision and recall across thresholds. The trend shows peak mAP within an optimal confidence range, indicating a balance between minimizing false positives and avoiding missed detections. Strengths include consistent passenger detection across frames. Limitations involve sensitivity to threshold settings, which may affect detection under extreme lighting or occlusion conditions.



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Chapter 5

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter presents the findings, conclusions, and recommendations derived from the analysis and interpretation of the results. The primary objective of the proposed system is to develop a non-contact overloading detection system for public utility buses using the YOLOv8 model for real-time passenger monitoring. The system is designed to accurately detect the number of passengers inside the vehicle through camera-based computer vision and automated data processing.

The system aims to support transportation safety and regulatory enforcement by providing an efficient and reliable method of identifying overloading conditions without requiring direct physical inspection. Through the integration of image processing, real-time monitoring, and raspberry pi 4, based data handling, the system seeks to improve operational accuracy, enhance monitoring capabilities, and contribute to safer public transportation practices.

Summary of Results

The four system test trials were conducted with actual passenger counts of 17, 12, 13, and 16 individuals, while the system detected 14, 8, 14, and 11, resulting in accuracies of 82%, 67%, N/A (over-detection), and 69%. Errors observed during testing included undetected individuals at the back, missed illustrated figures, over-counting in some frames, and reduced detection performance under varied lighting and crowded conditions. Survey responses reflected strong agreement that the system accurately detects passenger counts (Mean = 4.5), indicating that the computer vision module processes real-time video feeds reliably even with



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bus movement and changing lighting. The YOLOv8 model demonstrated accurate localization, fast detection speed, and clear visualization through bounding boxes, although limitations were present in cases of occlusion, poor lighting, camera angle issues, and full-capacity scenarios. Respondents also agreed that hardware components operate correctly (Mean = 4.46) and that the system remains reliable under different passenger loads (Mean = 4.34), confirming that the image processing pipeline performs effectively in dynamic environments. Additionally, the Raspberry Pi 4 integration was rated as simple and reliable (Mean = 4.32), as it consistently managed data flow, interpreted detections, and transmitted outputs without performance issues. Backend console logs showed stable processing of video frames, generation of overload alerts, and transmission of MQTT messages, with strengths including accurate counting, real-time responsiveness, and successful alert communication, alongside weaknesses related to video quality, lighting, crowd density, and network dependence.

The survey results indicate strong user validation of the system's alert transmission capabilities. Respondents strongly agreed that alerts are sent promptly to authorities (Mean = 4.48) and that the alert system remains reliable and consistent during operation (Mean = 4.44), showing confidence in the wireless communication module. Users also strongly agreed that the system automatically generates clear and usable digital records of overloading events, reflected in the usability rating for the alert interface (Mean = 4.42). The system's web-based dashboard was positively received, with respondents agreeing that it integrates smoothly into existing workflows (Mean = 4.36) and can be monitored remotely for real-time supervision (Mean = 4.34). Additionally, the Bus Alerts interface displays overload notifications with time stamps, detected passenger counts, and GPS coordinates, offering immediate and detailed



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information to personnel. The interface's strengths include real-time visibility of events, accurate passenger counts, precise location tracking, and complete timestamp logging, although its performance may be affected by GPS accuracy and network connectivity issues.

Survey results indicate that the majority of respondents rated the system positively for passenger detection accuracy in real-world conditions, with 27 strongly agreeing and 18 agreeing, yielding a mean score of 4.44 ("Strongly Agree"). Dependability was also highly rated: the system's reliable performance under normal bus operations received a mean of 4.40 ("Strongly Agree"), while consistency under peak passenger loads scored 4.34 ("Agree"). Alert responsiveness was evaluated positively as well, with 24 respondents strongly agreeing and 21 agreeing, resulting in a mean of 4.38 ("Strongly Agree"), reflecting confidence in the system's ability to transmit timely real-time data. Technical testing confirms these perceptions, showing that real-time monitoring achieves an average frame processing speed of 0.04–0.09 seconds, with accuracy rated at 4.19 ("Agree") and both reliability and responsiveness at 4.37 ("Strongly Agree"). Strengths of the system include fast processing speeds and stable operation under normal conditions, while limitations involve occasional network-dependent delays and reduced accuracy during periods of high passenger density.

The system demonstrated strong performance based on ISO/IEC 25010 quality characteristics. A total of 58% strongly agreed and 40% agreed that the system effectively detects overloading, counts passengers, and communicates with regulatory platforms. Reliability and efficiency both achieved weighted means of 4.37, with only minor slowdowns during extended operation. Security was rated highly (mean = 4.44), and usability achieved a mean of 4.46, reflecting positive user experiences. Maintainability also scored highly (54% strongly agree; 45% agree). Limitations included sensitivity to environmental conditions,



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occasional performance drops during long-duration use and suggested improvements in documentation and user interface clarity.

Survey results indicate strong agreement that the system can integrate with cloud platforms, with 28 respondents strongly agreeing and 17 agreeing, yielding a mean score of 4.46 (“Strongly Agree”). Respondents also rated the system’s adaptability for use in other public transportation modes positively, with a mean of 4.40 (“Strongly Agree”), reflecting confidence in its potential beyond bus operations. Scalability was assessed in terms of long-term system design and documentation support, receiving mean scores of 4.34 and 4.32, respectively (“Agree”), demonstrating that the system can support broader deployment and sustained monitoring. Performance evaluation of the YOLOv8 detection model showed high accuracy, precision, and recall, ensuring reliable passenger identification across frames. Metrics revealed strengths in stable precision and consistent detection, while limitations included trade-offs between precision and recall, reduced detection for partially obstructed or distant passengers, and sensitivity to extreme lighting or occlusion conditions. Overall, these results confirm that the system is capable of accurate, real-time monitoring and adaptable for integration with cloud platforms and expanded deployment.

Conclusion

The system demonstrates reliable real-time passenger monitoring supported by accurate detection, stable hardware performance, and dependable data processing through the Raspberry Pi 4, making it effective under typical operating conditions. While the YOLOv8 model and embedded components perform well across dynamic environments, the system’s accuracy is influenced by lighting variability, occlusion, camera placement, environmental movement, and network stability. These factors contributed to inconsistent detection results



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observed during system trials. Despite these limitations, the combined performance of the computer vision module, image processing pipeline, and embedded controller confirms that the system is practical and functional for real-time operations, although further optimization is necessary to improve detection consistency in crowded, low-light, and visually complex scenarios.

The findings show that the system provides a dependable and efficient alert transmission process, capable of delivering real-time notifications and digital records to support enforcement activities. Its high usability and seamless integration with existing workflows demonstrate that the dashboard and alert interface can be readily adopted by authorities for immediate monitoring and decision-making. The inclusion of accurate timestamps, passenger counts, and location data strengthens its functionality as a tool for regulatory compliance and operational awareness. While minor limitations such as GPS precision and network dependency may influence performance in certain conditions, the overall results indicate that the alert system is well-designed, responsive, and effective for supporting overloading detection and enforcement tasks.

The system demonstrates reliable and accurate real-time monitoring, maintaining consistent detection and alert responsiveness under varying operational conditions. While it performs efficiently in normal operations, its performance can be slightly affected by network delays and crowded boarding scenarios. Nonetheless, the system's high ratings for accuracy, reliability, and responsiveness indicate that it effectively supports passenger detection and timely alert transmission in practical bus environments.

The system meets the functional requirements for detecting passenger overloading and communicating with regulatory platforms. Users confirm that the system performs core



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functions accurately, though attention to environmental sensitivity is necessary. The system is reliable and efficient for real-time monitoring and can sustain performance in varying operational conditions. Improvements in long-duration operation may enhance its overall reliability. The system provides a secure environment for data storage and transmission. Strengthening authentication measures would further enhance security. The system is highly usable and accessible to operators. Minor interface refinements could further improve user experience and readability. The system is maintainable and adaptable for future upgrades. More comprehensive documentation would support smoother maintenance and development.

The system demonstrates strong potential for integration with cloud-based monitoring and external databases, as well as adaptability to other public transportation settings. Its high accuracy, precision, and recall in passenger detection confirm its suitability for real-time overloading monitoring. Although performance can be affected under extreme lighting, occlusions, or high-confidence thresholds, the system is generally robust, scalable, and capable of supporting long-term monitoring and centralized data management for broader deployment.

Recommendation

Based on test results and limitations discovered in the system testing and evaluation processes involving the Non-Contact Overloading Detection System for Public Utility Buses Using the YOLOv8 Algorithm, recommendations have been made to enable improvements in terms of system effectiveness, dependability, and scalability. These recommendations have been made with respect to improvements in detection precision, system performance capabilities using hardware resources, alert system designs, real-world implementations, and system quality according to the ISO/IEC 25010 standard definitions.



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1. System Accuracy and Hardware Performance: Future iterations of the system may improve YOLOv8 training by incorporating additional datasets that include varied lighting conditions, different crowd densities, and multiple camera angles. Adjusting camera placement, using wider-angle or higher-resolution lenses, and optimizing onboard lighting or applying low-light image enhancement techniques can further reduce occlusion and improve detection accuracy. Upgrading thermal management and refining the Raspberry Pi processing pipeline may also help sustain real-time performance during prolonged operation, ensuring stability even under high passenger loads.

2. Alert System and Dashboard: The alert system can be enhanced by implementing backup communication channels, such as SMS fallback or dual-network setups, to maintain performance during weak connectivity. GPS accuracy may be improved through assisted GPS integration or multi-source geolocation for more precise vehicle tracking. Additionally, incorporating real-time system health monitoring would help prevent failures in automated logging, ensuring uninterrupted generation of digital reports and alerts to authorities.

3. Real-World Performance: Future research should focus on calibrating detection confidence thresholds to achieve a balance between accuracy and false detections, particularly in crowded conditions. Optimizing detection speed by selecting appropriate YOLOv8 model sizes (e.g., YOLOv8n or YOLOv8s) tailored for bus environments can enhance system responsiveness. More extensive field testing is recommended under varied passenger movement speeds and peak-hour conditions to evaluate real-world performance and identify further areas for optimization.

4. ISO 25010 Quality Enhancements: System quality can be improved by implementing optional two-factor authentication (2FA) for stronger dashboard security. User



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interface readability may be enhanced by enlarging text, increasing contrast, and refining button labels to support quick comprehension during operation. Comprehensive documentation for maintenance, updates, and troubleshooting is recommended to facilitate system upkeep. Integrating automated recovery and reboot mechanisms would ensure long-duration stability, maintaining efficiency during continuous operation.

5. Future Development and Scalability: To enable broader deployment, the system should include cloud-based storage and centralized monitoring modules capable of supporting multiple buses and cities. Pilot testing on other types of public transport, such as jeepneys, vans, and provincial buses, with tailored camera calibration is recommended to verify adaptability. National-level deployment may require server load testing and network optimization to ensure stable operation. Finally, enhancing the YOLOv8 model's robustness through additional training samples that cover occlusions, backlighting, and standing passengers will improve detection accuracy under diverse operational conditions.



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APPENDICES



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Appendix A LETTER FROM LAND TRANSPORT OFFICE (LTO)

 CITY GOVERNMENT OF CALAMBA
CITY COLLEGE OF CALAMBA
Dalubhasaan ng Lungsod ng Calamba
ALCUOA Level I Accredited
Global Citizenship Education - Regional Hub

Department of Computing and Informatics

April 25, 2025

Land Transport Office (LTO)
City Of Calamba

Dear Ma'am / Sir

Good day! We, the third-year Computer Science Students from City College of Calamba, are conducting research for our **Thesis 1**. We are currently conducting a research study titled " **Non-Contact Overloading Detection System for Public Utility Buses Using YOLO v8 Algorithm** " as part of our academic requirements. In line with this, we respectfully request permission to conduct a survey within your organization, community to gather essential information and insights related to our study.

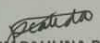
Rest assured that all data collected will be treated with strict confidentiality and used solely for academic purposes.

Should you require further details, please do not hesitate to contact us. We look forward to your positive response

Thank you for your time and support.

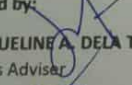
Sincerely,

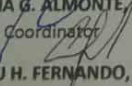
JOHN ARNIE BARBA
Researcher

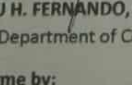

ANN PAULINA P. CALIDA
Researcher


LEMUEL G. DICKE
Researcher


DHANE KIAN G. HELLIE
Researcher

Noted by:

JACQUELINE A. DELA TORRE, PhD
Thesis Adviser


REGINA G. ALMONTE, PhD
Thesis Coordinator


ARLOU H. FERNANDO, MM-ITM
Dean, Department of Computing and Informatics

Conforme by:
LAND TRANSPORT OFFICE (LTO)
City of Calamba

 Old Municipal Site, Brgy. VII, Poblacion, Calamba City, Laguna
(049) 559-8900 to 8907 / (02) 8-5395-17



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Appendix B SIGNED EXPERT VALIDATION LETTERS

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Global Citizenship Education - Regional Hub

Department of Computing and Informatics

October 24, 2023

Prof. Jonimari Ross Valencia
Faculty
City College of Calamba
Calamba City, Laguna

Dear **Prof. Valencia**:

Good day!

We, the undersigned group of researchers, pursuing a Bachelor's degree in Computer Science, we are conducting a thesis entitled "Non-Contact Overloading Detection System for Public Utility Buses Using Yolov8 Algorithm." Our project's primary objective is **Safety and Compliance focus**, to contribute to commuter safety by creating an automated overloading detection system that ensures compliance with legal passenger capacity limits for public utility buses.

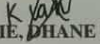
Recognizing your esteemed expertise, we are reaching out to respectfully request your participation as one of our distinguished validators of our developed system and research instrument. Your valuable insights and knowledge would be immensely beneficial to our project's advancement and ultimate success. Thank you so much!

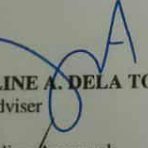
Respectfully yours,

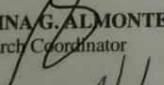

BARBA, JOHN ARNIE


DICHE, LEMUEL G.

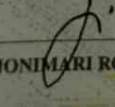

CALIDA, ANN PAULINA P.


HELLIE, PHANE KIAN

Noted by: 
JACQUELINE A. DELA TORRE, PhD
Research Adviser

Recommending Approval:

REGINA G. ALMONTE, PhD
Research Coordinator

Approved by: 
ARLOU H. FERNANDO, MM-ITM
Dean

Conforme:

JONIMARI ROSS VALENCIA

Old Municipal Site, Brgy. VII, Poblacion, Calamba City, Laguna
(049) 559-8900 to 8907 / (02) 8-5395-170 to 171



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Department of Computing and Informatics

October 24, 2023

Dr. Rowan N. Elomina
Faculty
City College of Calamba
Calamba City, Laguna

Dear Mr. Elomina:

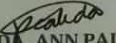
Good day!

We, the undersigned group of researchers, pursuing a Bachelor's degree in Computer Science, we are conducting a thesis entitled "Non-Contact Overloading Detection System for Public Utility Buses Using Yolov8 Algorithm." Our project's primary objective is **Safety and Compliance focus**, to contribute to commuter safety by creating an automated overloading detection system that ensures compliance with legal passenger capacity limits for public utility buses.

Recognizing your esteemed expertise, we are reaching out to respectfully request your participation as one of our distinguished validators of our developed system and research instrument. Your valuable insights and knowledge would be immensely beneficial to our project's advancement and ultimate success. Thank you so much!

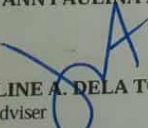
Respectfully yours,


BARBA, JOHN ARNIE

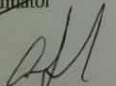

CALIDA, ANN PAULINA P.

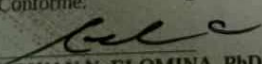

DICHE, LEMUEL G.


HELLIE, DHANE KIAN

Noted by: 
JACQUELINE A. DELA TORRE, PhD
Research Adviser

Recommending Approval:
REGINA C. ALMONTE, PhD
Research Coordinator

Approved by: 
ARLOU H. FERNANDO, MM-ITM
Dean

Conforme: 
ROWAN N. ELOMINA, PhD





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Department of Computing and Informatics

October 24, 2023

Prof. Nimfa S. Racelis
Faculty
City College of Calamba
Calamba City, Laguna

Dear **Prof. Racelis**:

Good day!

We, the undersigned group of researchers, pursuing a Bachelor's degree in Computer Science, we are conducting a thesis entitled "**Non-Contact Overloading Detection System for Public Utility Buses Using Yolov8 Algorithm.**" Our project's primary objective is **Safety and Compliance focus**, to contribute to commuter safety by creating an automated overloading detection system that ensures compliance with legal passenger capacity limits for public utility buses.

Recognizing your esteemed expertise, we are reaching out to respectfully request your participation as one of our distinguished validators of our developed system and research instrument. Your valuable insights and knowledge would be immensely beneficial to our project's advancement and ultimate success. Thank you so much!

Respectfully yours,

BARBA, JOHN ARNIE
CALIDA, ANN PAULINA P.

DICHE, LEMUEL G.
HELLIE, DHANE KIAN

Noted by:
JACQUELINE A. DELA TORRE, PhD
Research Adviser

Recommending Approval:
REGINA G. ALMONTE, PhD
Research Coordinator

Approved by:
ARLOU H. FERNANDO, MM-ITM
Dean

Conformer:
NIMFA S. RACELIS, MIT

Old Municipal Site, Brgy. VII, Poblacion, Calamba City



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Appendix C CERTIFICATE FOR INSTRUMENT AND SYSTEM VALIDATORS



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Department of Computing and Informatics

CERTIFICATE OF INSTRUMENT VALIDATION

This is to certify that I have validated the interview protocol of the study entitled

“Non-Contact Overloading Detection System for Public Utility Buses using Yolov8 Algorithm”

Prepared by

Barba, John Arnie
Calida, Ann Paulina P.
Diche, Lemuel G.
Hellie, Dhane Kian G.

Name of the Researcher

and I have found it complete and satisfactory in terms of face and content validity.

Rowan N. Elomina, PhD

Name of Validator

Faculty, Department of Computing and Informatics

Affiliation

October 24, 2025

Date



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Name of the Researcher

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Jonimari Ross Valencia

Name of Validator

Faculty, Department of Computing and Informatics

Affiliation

October 24, 2025

Date



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Continuation of Appendix C...



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Calida, Ann Paulina P.
Diche, Lemuel G.
Hellie, Dhane Kian G.

Name of the Researcher

and I have found it complete and satisfactory in terms of face and content validity.

Nimfa S. Racelis, MIT

Name of Validator

Faculty, Department of Computing and Informatics

Affiliation

October 24, 2025

Date



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CERTIFICATE OF SOFTWARE VALIDATION

This is to certify that I have validated the interview protocol of the study entitled

"Non-Contact Overloading Detection System for Public Utility Buses using Yolov8 Algorithm"

Prepared by

Barba, John Arnie
Calida, Ann Paulina P.
Diche, Lemuel G.
Hellie, Dhane Kian G.

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Appendix D SURVEY INSTRUMENT USING ISO 25010



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SOFTWARE EVALUATION FORM

Name(optional):

Purpose: This survey evaluates the **NON-CONTACT OVERLOADING DETECTION SYSTEM FOR PUBLIC UTILITY BUSES USING YOLO v8 ALGORITHM** the ISO/IEC 25010 quality model.

Instruction: Please rate each statement based on your level of agreement by putting a check mark on the rows column that corresponds to the assessment.

Scale: 1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

1. To create and build a hardware-based system that can detect overloading based on:

	1	2	3	4	5
1. The System accurately detects passenger count					
2. Hardware components function properly					
3. System operates reliably under bus conditions					
4. System works consistently under varying loads					
5. Easy to implement and integrate					

2. To establish an alert system that automatically notifies concerned stakeholders in case of overloading in terms of:

	1	2	3	4	5
1. Alerts are transmitted promptly to authorities					
2. Alert system is reliable and consistent					
3. Alert interface is easy to use					
4. System integrates smoothly with existing workflow					
5. System can be monitored remotely					



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3. To evaluate the system's performance in real-world public transportation settings in terms of:

	1	2	3	4	5
1. System accurately detects passengers in real-world conditions					
2. System performs reliably under bus operation					
3. Alert response is prompt and effective					
4. System maintains consistency under peak loads					
5. System is adaptable to different bus routes					

4. To determine the system's level of effectiveness as evaluated through the ISO 25010 quality standards in terms of:

A. Functional Suitability and Compatibility

Indicators		1	2	3	4	5
Functional Completeness	The system provides all the necessary functions for detecting passenger overloading.					
Functional Correctness	The system accurately counts passengers and determines when capacity is exceeded.					
Functional Appropriateness	The system functions are appropriate and relevant for real-time bus monitoring.					
Co-existence	The system operates smoothly alongside other onboard bus devices (e.g., cameras, sensors).					
Interoperability	The system can communicate or share data with transportation regulatory platforms.					



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B. Security and Reliability

Indicators		1	2	3	4	5
Confidentiality	The system protects stored and transmitted passenger information from unauthorized access.					
Recoverability	The system can recover quickly after errors or interruptions.					
Integrity	The system prevents alteration or tampering of detection results.					
Accountability	The system maintains authenticity and integrity of collected data.					
Faultlessness	The system functions consistently under different conditions.					

C. Reliability, Performance Efficiency

Indicators		1	2	3	4	5
Availability	The system is available and operational when needed.					
Fault Tolerance	The system can handle minor faults without failing completely.					



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Time Behavior	The system processes detection results quickly without noticeable delay.					
Resource Utilization	The system performs efficiently without excessive use of hardware resources.					
Capacity	The system can handle large passenger volumes without performance degradation.					

D. Usability, Flexibility

Indicators		1	2	3	4	5
Appropriateness Recognizability	The system's interface is easy to recognize and understand.					
Learnability	The system is easy to learn and operate, even for new users.					
Operability	The system provides clear feedback or error messages to guide errors.					
User Error Protection	The system promotes user engagement and accessibility for all operators.					
Installability	The system promotes user The system's performance scales well with increased passenger volume.					



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E. Maintainability, Flexibility, Safety

Indicators		1	2	3	4	5
Modularity	The system is designed so components can be maintained or replaced easily.					
Reusability	The system allows certain functions or components to be reused for future improvements or related systems.					
Operational Constraint	The system operates safely under various environmental conditions.					
Risk Identification	The system identifies risks that could affect passenger safety.					
Hazard Warning	The system provides alerts or warnings in case of malfunction.					

5. To suggest system enhancements and future development directions with emphasis on:

	1	2	3	4	5
1. The System can integrate with the cloud platforms					
2. System allows for future upgrades and enhancements					
3. Interface supports easy adaptation to other transport					
4. System design is scalable for long-term use					
5. Documentation supports maintainability					



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Appendix E LETTER FOR CONSENT OF RESPONDENTS



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October 27, 2025

City Of Calamba

Dear Respondents,

Good day! We, the fourth-year Computer Science Students from City College of Calamba, we are 4th-year Computer Science students currently developing a system called the Non-Contact Overloading Detection System for Public Utility Buses Using the YOLOv8 Algorithm. As part of our research, we are conducting a comparative analysis of data gathered through our system and traditional manual passenger counting methods.

This initiative aims to improve the monitoring and enforcement of passenger capacity regulations in public transportation, ensuring safer and more efficient bus operations. Our goal is to create an accurate, automated, and reliable system that supports regulatory agencies and transport operators in maintaining commuter safety and promoting compliance with overloading policies.

Rest assured that all data collected will be treated with strict confidentiality and used solely for academic purposes. We look forward to your positive response.

Thank you for your time and support.

Sincerely,

John Arnie Barba
JOHN ARNIE BARBA
Researcher

Ann Paulina P. Calida
ANN PAULINA P. CALIDA
Researcher

Lemuel G. Diche
LEMUEL G. DICHE
Researcher

Dhans Kian G. Hellie
DHANS KIAN G. HELLIE
Researcher

Noted by:

Jacqueline A. Dela Torre
JACQUELINE A. DELA TORRE, PhD
Thesis Adviser

Regina G. Almonte
REGINA G. ALMONTE, PhD
Thesis Coordinator

Arlo H. Fernando
ARLOU H. FERNANDO, MM-ITM
Dean, Department of Computing and Informatics

Conforme by:

Respondents
City of Calamba



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Appendix F INTERVIEW AND SURVEY QUESTIONNAIRES



NON-CONTACT OVERLOADING DETECTION SYSTEM FOR PUBLIC UTILITY BUSES USING YOLO v8 ALGORITHM

We are 4th-year Computer Science students currently developing a system called the Non-Contact Overloading Detection System for Public Utility Buses Using the YOLOv8 Algorithm.

As part of our research, we are conducting a comparative analysis of data gathered through our system and traditional manual passenger counting methods. This initiative aims to improve the monitoring and enforcement of passenger capacity regulations in public transportation, ensuring safer and more efficient bus operations.

Our goal is to create an accurate, automated, and reliable system that supports regulatory agencies and transport operators in maintaining commuter safety and promoting compliance with overloading policies.



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NON-CONTACT OVERLOADING DETECTION SYSTEM FOR PUBLIC UTILITY BUSES USING YOLO v8 ALGORITHM

"Data will be gathered through a questionnaire to capture demographics, gaming habits, and perceptions of learning outcomes. Internet research will supplement insights from existing literature and trends. Observations in school and computer shop settings will provide additional context on user engagement and learning behaviors. Ethical considerations include obtaining permission and ensuring confidentiality. These methods aim to comprehensively understand user interaction with the computer science learning game module."

In accordance of The Republic Act 10173, officially known as the Data Privacy Act of 2012 (DPA), is Philippine's data privacy law, aiming to "to protect the fundamental human right of privacy, of communication while ensuring free flow of information to promote innovation and growth" while also ensuring "that personal information in information and communications systems in the government and in the private sector are secured and protected."

Personal Information

In accordance of The Republic Act 10173, officially known as the [Data Privacy Act of 2012](#) (DPA), is Philippine's data privacy law, aiming to "to protect the fundamental human right of privacy, of communication while ensuring free flow of information to promote innovation and growth" while also ensuring "that personal information in information and communications systems in the government and in the private sector are secured and protected."

Name (optional)

Your answer

Gender *

☐ Male

☐ Female

Qualification *

☐ Passenger

☐ Authorities

Age *

Your answer



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Instruction: Please rate each statement based on your level of agreement by putting a check mark on the rows column that corresponds to the assessment.

Scale: 1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

1. To create and build a hardware-based system that can detect overloading based on:

Hardware Detection

Hardware Detection *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. The System accurately detects passenger count	☐	☐	☐	☐	☐
2. Hardware components function properly	☐	☐	☐	☐	☐
3. System operates reliably under bus conditions	☐	☐	☐	☐	☐
4. System works consistently under varying loads	☐	☐	☐	☐	☐
5. Easy to implement and integrate	☐	☐	☐	☐	☐



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2. To establish an alert system that automatically notifies concerned stakeholders in case of overloading in terms of:

Alert System

Alert System *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. Alerts are transmitted promptly to authorities	☐	☐	☐	☐	☐
2. Alert system is reliable and consistent	☐	☐	☐	☐	☐
3. Alert interface is easy to use	☐	☐	☐	☐	☐
4. System integrates smoothly with existing workflow	☐	☐	☐	☐	☐
5. System can be monitored remotely	☐	☐	☐	☐	☐



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3. To evaluate the system's performance in real-world public transportation settings in terms of:

System Performance

System Performance *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. System accurately detects passengers in real-world conditions	☐	☐	☐	☐	☐
2. System performs reliably under bus operation	☐	☐	☐	☐	☐
3. Alert response is prompt and effective	☐	☐	☐	☐	☐
4. System maintains consistency under peak loads	☐	☐	☐	☐	☐
5. System is adaptable to different bus routes	☐	☐	☐	☐	☐



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4. To determine the system's level of effectiveness as evaluated through the ISO 25010 quality standards in terms of:

Functional Suitability *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The system provides all the necessary functions for detecting passenger overloading.	☐	☐	☐	☐	☐
The system accurately counts passengers and determines when the capacity is exceeded	☐	☐	☐	☐	☐
The system functions are appropriate and relevant for real-time bus monitoring.	☐	☐	☐	☐	☐

The system operates smoothly alongside other onboard bus devices (e.g., cameras, sensors).	☐	☐	☐	☐	☐
The system can communicate or share data with transportation regulatory platforms.	☐	☐	☐	☐	☐



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Security *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The system protects stored and transmitted passenger information from unauthorized access.	☐	☐	☐	☐	☐
The system can recover quickly after errors or interruptions.	☐	☐	☐	☐	☐
The system prevents alteration or tampering of detection results.	☐	☐	☐	☐	☐
The system maintains authenticity and integrity of collected data.	☐	☐	☐	☐	☐
The system functions consistently under different conditions.	☐	☐	☐	☐	☐



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Performance Efficiency *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The system is available and operational when needed.	☐	☐	☐	☐	☐
The system can handle minor faults without failing completely.	☐	☐	☐	☐	☐
The system processes detection results quickly without noticeable delay.	☐	☐	☐	☐	☐
The system performs efficiently without excessive use of hardware resources.	☐	☐	☐	☐	☐
The system can handle large passenger volumes without performance degradation.	☐	☐	☐	☐	☐
The system can handle large passenger volumes without performance degradation.	☐	☐	☐	☐	☐



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Usability *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The system's interface is easy to recognize and understand.	☐	☐	☐	☐	☐
The system is easy to learn and operate, even for new users.	☐	☐	☐	☐	☐
The system provides clear feedback or error messages to guide errors.	☐	☐	☐	☐	☐
The system promotes user engagement and accessibility for all operators.	☐	☐	☐	☐	☐
The system promotes user The system's performance scales well with increased passenger volume.	☐	☐	☐	☐	☐



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Maintainability *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The system is designed so components can be maintained or replaced easily.	☐	☐	☐	☐	☐
The system allows certain functions or components to be reused for future improvements or related systems.	☐	☐	☐	☐	☐
The system operates safely under various environmental conditions.	☐	☐	☐	☐	☐
The system identifies risks that could affect passenger safety.	☐	☐	☐	☐	☐
The system provides alerts or warnings in case of malfunction.	☐	☐	☐	☐	☐



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5. To suggest system enhancements and future development directions with emphasis on:

System Enhancements

System Enhancements *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. The System can integrate with the cloud platforms	☐	☐	☐	☐	☐
2. System allows for future upgrades and enhancements	☐	☐	☐	☐	☐
3. Interface supports easy adaptation to other transport	☐	☐	☐	☐	☐
4. System design is scalable for long-term use	☐	☐	☐	☐	☐
5. Documentation supports maintainability	☐	☐	☐	☐	☐



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Appendix G LANGUAGE EDITOR CERTIFICATION



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Appendix H USER MANUAL



Non-Contact Overloading Detection System for Public Utility Buses Using YOLOv8 Algorithm

Developers:

Dhane Kian G. Hellie

John Arnie Barba

Ann Paulina P. Calida

Lemuel G. Diche

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Troubleshooting

5.1. Common Issues and Solutions

Frequently Asked Questions (FAQs)

Contact Information

Introduction

The Bus Overload Detection Alert System is a mobile application designed to assist transport authorities in monitoring public utility buses for passenger overload incidents in real time.

The app connects to a backend server that collects data from the detection hardware (Raspberry Pi, camera, and GPS). Once an overload is detected, the system automatically sends alerts and notifications to the app.

The application also allows users to view a list of overload alerts, check details such as bus number, time, and status, and locate the detected bus on Google Maps.

2. System Overview

The app integrates three core components:

- Raspberry Pi-based Detection Device – detects overloaded buses using sensors and sends data to the backend.**
- MQTT Protocol – transfers detection data to the mobile app.**
- Mobile Application – displays alerts to the mobile app**

Main Features:

- Live detection via MQTT**
- Automatic push notifications even when the app is closed**
- Real-time alert list with bus details**
- “Locate” feature using Google Maps**
- Exit button for convenient app closing**

System Overview

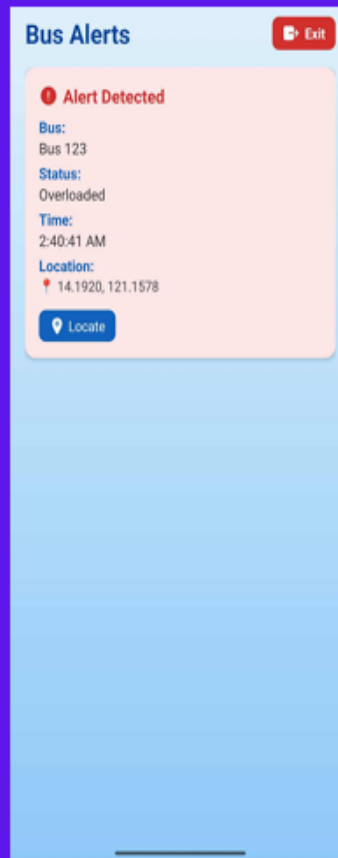


- **Raspberry Pi-based Detection Device** – detects overloaded buses using sensors and sends data to the backend.

System Overview



- **MQTT Protocol** – transfers detection data to the mobile app.



2. System Overview

- **Mobile Application – displays alert to the mobile app**

Main Features

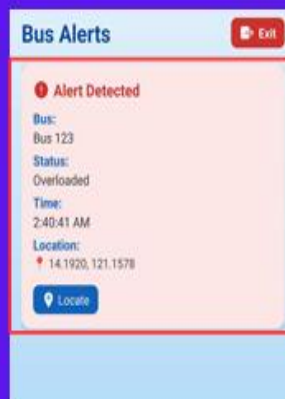
- **Live detection via MQTT**
- **Real-time alert list with details**
- **“Locate” feature to open bus location on Google Maps**
- **Exit button for convenient app closing**

Live detection via MQTT



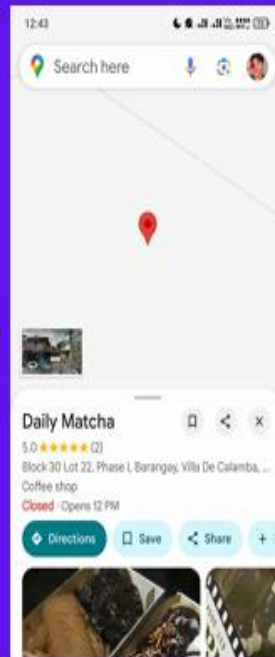
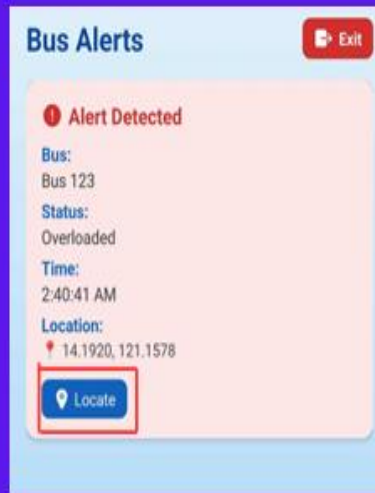
The Live Detection via MQTT feature allows the app to receive real-time alerts from the system whenever a bus is detected as overloaded. The app connects to an MQTT broker (Message Queuing Telemetry Transport), which acts as a communication bridge between the backend system and the mobile application. When the backend detects an overloaded bus, it sends a message containing details such as the bus number, status, time, and location coordinates to the MQTT topic. Once the message is published, the alert instantly appears on the app's screen

Real-time alert list with details



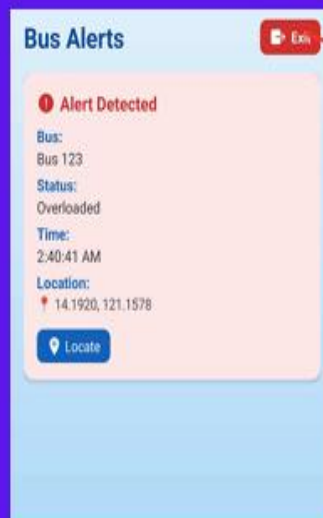
The Real-time Alert List continuously monitors and displays bus overload incidents as they happen. It works through a combination of MQTT messaging, backend detection, and mobile app updates. When the backend system detects an overloaded bus, it sends a data message to an MQTT broker. The mobile app subscribes to the specific topic (bus/overload) on this broker and immediately receives the alert message once published.

“Locate” feature to open bus location on Google Maps



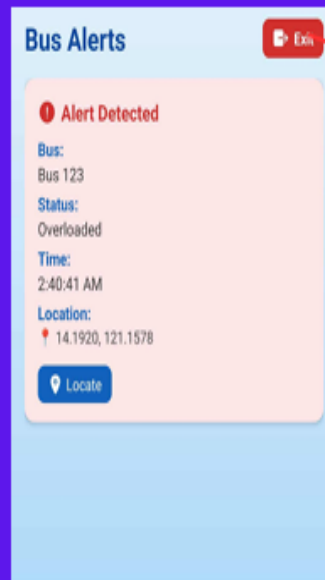
The “Locate” feature allows users to view the exact location of an overloaded bus directly on Google Maps. Each alert received by the app contains the latitude and longitude coordinates of the detected bus, which are transmitted from the backend system during detection. When the user taps the “Locate” button in the alert details, the app automatically opens Google Maps and points to the specific coordinates of that bus.

Exit button for convenient app closing



The Exit button provides users with an easy and secure way to close the application directly from within the interface. When the user taps the “Exit” button, the app displays a confirmation message asking if they are sure they want to exit. This step prevents accidental closure while using the app.

Exit button for convenient app closing



The Exit button provides users with an easy and secure way to close the application directly from within the interface. When the user taps the "Exit" button, the app displays a confirmation message asking if they are sure they want to exit. This step prevents accidental closure while using the app.

3. Installation Guide

3.1. Requirements

- **Android phone (version 8.0 or above)**
- **Active internet connection**
- **Installed app (Bus Overload Detection Alert System)**

Android phone (version 8.0 or above)



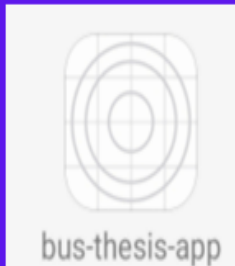
The Android phone (version 8.0 or above) serves as the primary device for running the Bus Alert Monitoring App. Android 8.0 (Oreo) and later versions provide the necessary system compatibility, security features, and performance enhancements required for smooth app operation.

Active internet connection



The application requires an active internet connection (Wi-Fi or mobile data) to function properly. It uses the internet to connect with the MQTT broker, which transmits real-time detection data from the backend device (Raspberry Pi). This connection allows the app to receive live alerts and open Google Maps for bus location tracking.

Installed app (Bus Overload Detection Alert System)



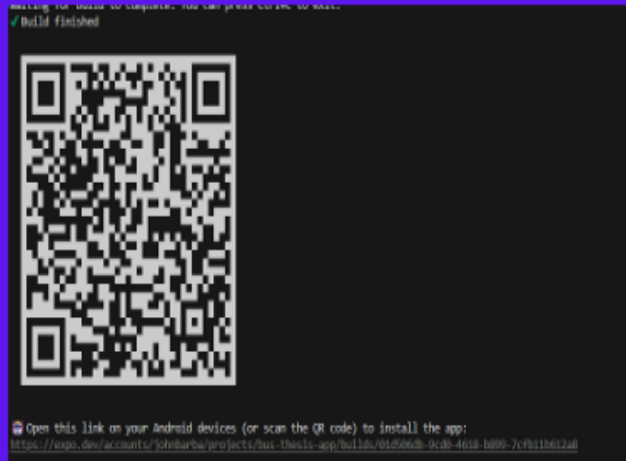
Android internal distribution build						
4ad9e79 · Initial commit						
Profile	Environment	SDK version	Version	Fingerprint	Commit	Created by
preview	preview	54.0.0	1.0.0 (10)	84df1ae	4ad9e79*	johnbarba
Build artifact APK						
Status	Start time	Wait time	Queue time	Build time	Total time	Availability
Finished	Nov 4, 2025 1:57 PM	0s	49s	17m 18s	18m 7s	12 days

The **Bus Overload Detection Alert System** mobile application must be properly installed on the user's Android device. Once installed, it allows transport authorities to monitor buses in real time and receive instant alerts whenever an overloaded bus is detected by the backend system.

3.2 Installation Steps

- **Obtain the APK file provided by the developers.**
- **Tap on the APK file to install.**
- **When prompted, allow installation from unknown sources (if necessary).**
- **Once installed, open the app.**

Obtain the APK file provided by the developers.



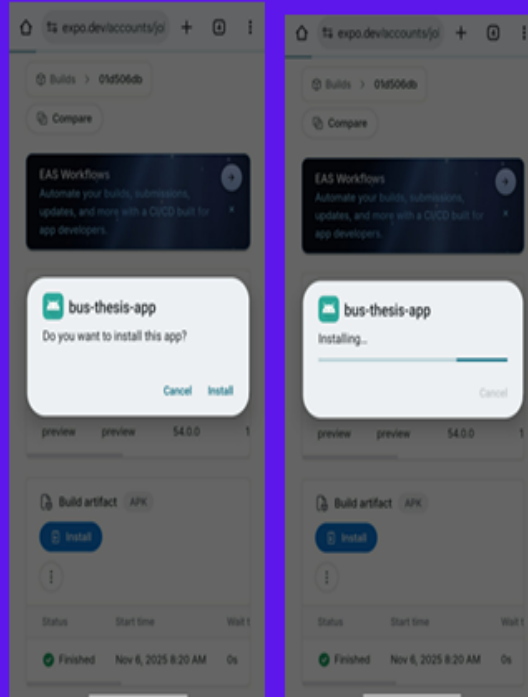
The Expo Dev Build process allows developers to create a customized Android APK version of the application that can be easily shared and installed. Once the build process is completed, Expo automatically generates a QR code that links directly to the APK file.

Tap on the APK file to install.

Build artifact APK							Install	Open with Orbit	⋮
Status	Start time	Wait time	Queue time ⌚	Build time	Total time	Availability ⌚			
Finished	Nov 6, 2025 8:20 AM	0s	1m 1s	9m 9s	10m 10s	11 days			

After accessing the APK file, the user simply needs to tap on it to start the installation process. The device will automatically detect it as an Android application package and display an installation prompt.

When prompted, allow installation from unknown sources (if necessary).



During the installation process, the system may prompt the user to allow installation from unknown sources. This occurs because the APK file is not obtained from the Google Play Store, and Android devices are designed to restrict such installations by default for security purposes.

Allowing installation from unknown sources grants temporary permission for the file manager or browser used to open the APK to proceed with the installation. Once this permission is enabled, the device recognizes the APK as safe for installation, allowing the setup process to continue and successfully install the application.

4. Using the Application

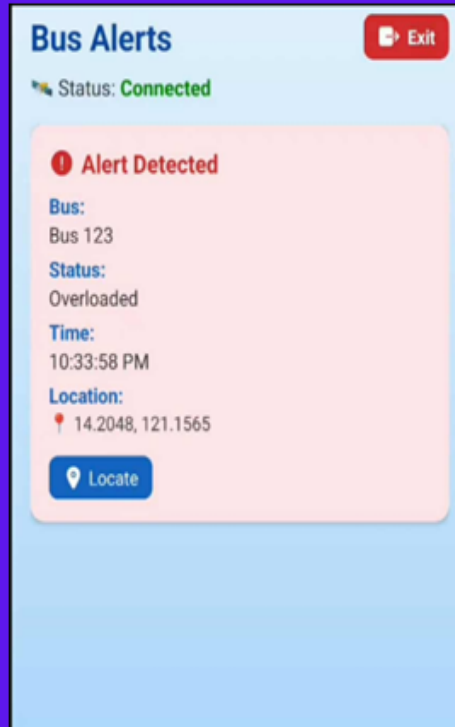
4.1. Viewing Alerts

Open the application.
Wait for the system to connect to the MQTT broker.

The Status will show "● Connected" when active.



- If it shows "● Offline," ensure your internet connection is stable. Once connected, any overload detection from buses will appear in the alert list.

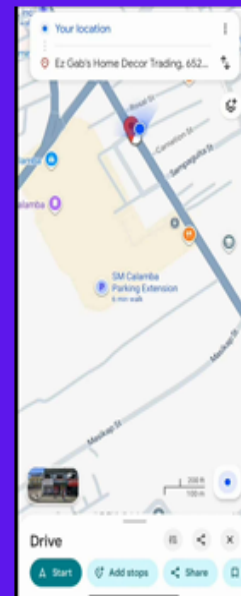
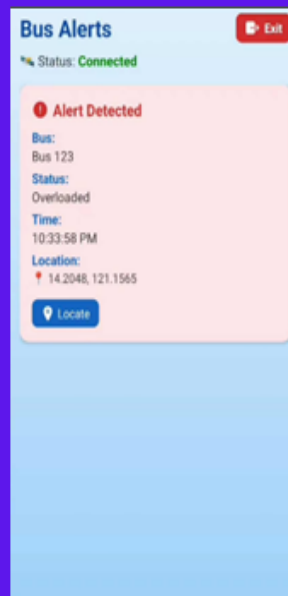


Each alert displays:

- **Bus Number**
- **Status**
- **Time**
- **Location**

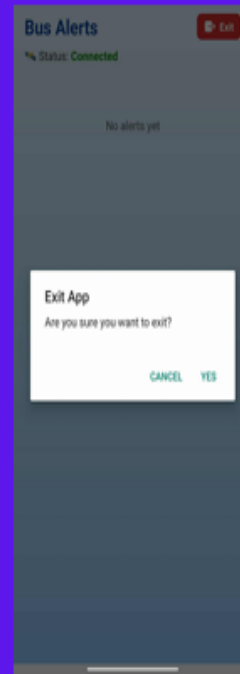
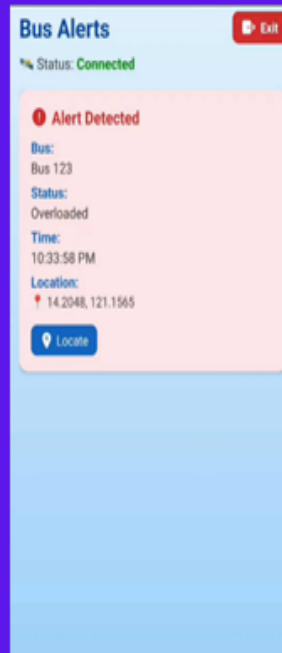
4.3. Locating the Bus

- In the alert list, tap the **Locate** button.
- The app will redirect you to Google Maps, showing the bus's exact coordinates.



4.4. Exiting the App

- Tap the Exit button located at the top-right of the screen.
- Confirm by selecting Yes on the dialog box.



Issue	Possible Cause	Solution
No alerts appear	Raspberry Pi or MQTT broker offline	Verify that the detection device and broker are active
App shows "Offline" status	No internet connection	Connect to Wi-Fi or mobile data
Location unavailable	GPS disabled or not detected	Check if GPS is active in the detection device

FAQs

Q1: Do I need to refresh the app manually?

A: Yes. The app manually updates alerts in real time through MQTT. So when the app is closed, no alerts will get received

Q2: What happens if the internet is disconnected?

A: The app will display an “Offline” status until the connection is restored.



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CURRICULUM VITAE



CITY COLLEGE OF CALAMBA

DEPARTMENT OF COMPUTING AND INFORMATICS Bachelor of Science in Computer Science

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Dhane Kian G. Hellie

Villa de Calamba, Calamba City, Laguna, 4027
Contact: 09271287536 | Email: kianhellie00@gmail.com

SUMMARY

Motivated and adaptable individual with a strong willingness to learn and take on new challenges. Able to work well with others, manage responsibilities effectively, and stay focused on achieving goals.

WORK EXPERIENCE

Production Operator

Continental Autonomous Mobility Phils.
Brgy. La Mesa, Calamba City, Laguna, Philippines November
2021 – May 2022

- Operated production machinery following safety protocols to maintain a safe working environment.
- Assisted in assembling products according to specifications and quality standards.
- Monitored production processes to ensure efficiency and adherence to schedules.

EDUCATION

Primary

Halang Elementary School (2009 – 2015)

Secondary

Calamba Institute (2015 – 2021)

Tertiary

City College of Calamba – Bachelor of Science in Computer Science (Expected Graduation: June 2026)

SKILLS

- | | |
|------------------------|---------------------------------------|
| • Communication Skills | * Microsoft Office (Word & Excel) |
| • Problem-Solving | * Google Docs & Google Sheets |
| • Teamwork | * Basic Computer Troubleshooting |
| • Time Management | * Basic Programming/Coding |
| • Adaptability | * File Management & Data Organization |

CHARACTER REFERENCES

Coun. Felix Mercado

Brgy. Counselor of Brgy. Lamesa Calamba City
09269107753

Adrianne Ortiz

RND.LPT
Senior High School Teacher, Nutritionist
09055221466

I hereby certify that the above stated information is true and correct.


DHANE KIAN G. HELLIE
Applicant ☐ s Complete Name with Signature



CITY COLLEGE OF CALAMBA

DEPARTMENT OF COMPUTING AND INFORMATICS
Bachelor of Science in Computer Science

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JOHN ARNIE BARBA

STUDENT

OBJECTIVE

☎ +63950-463-7153

✉ johnarniebarba12@gmail.com

📍 Ipil-Ipil Street, Barangay
Saimsim Calamba City, 4027

A Computer Science student seeking an internship opportunity to enhance my technical skills, gain practical experience, and assist in developing efficient and user-friendly applications. Committed to learning industry practices and supporting the team through quality work.

EDUCATION

City College of Calamba

2022 - Present

- Bachelor of Science in Computer Science
- Calamba, Laguna

Tanauan Institute Inc.

2020 - 2022

- Science, Technology, Engineering, and Mathematics
- Tanauan, Batangas

Sta. Anastacia - San Rafael National Highschool

2016-2020

- Junior High School
- Sto. Tomas, Batangas

Jose R. Platon Memorial School

2010-2016

- Primary School
- Calamba, Laguna

Character Reference

Regina G. Almonte, PhD

CCC Vice President for
Research, Extension, Planning
and Quality Assurance
rgalmonte@ccc.edu.ph
09399276321

TECHNICAL SKILLS

- Programming: Python, Java, JavaScript
- Website: HTML, CSS, JS
- MS Office: Excel, Word
- Database: XAMPP
- Tools: Visual Studio Code, Git/GitHub Copilot, PyCharm, NetBeans
- Editing: Canva, GIMP

OTHER SKILLS

- Excellent Time Management Skills
- Analytical with excellent creative problem solving skills
- Strong Interpersonal and Communication Skills

STRENGTHS

- Patient and hardworking
- Eager to learn new skills

I hereby certify that the information above is true and correct with the best of my knowledge and ability



CITY COLLEGE OF CALAMBA

DEPARTMENT OF COMPUTING AND INFORMATICS Bachelor of Science in Computer Science

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ANN PAULINA P. CALIDA

✉ calidapaulina@gmail.com
☎ 09053701042/09915962538

🏠 8 Don Bosco St. Ma. Leonora
Subd. Mayapa, Calamba City
Laguna

Education

BACHELOR OF SCIENCE IN COMPUTER SCIENCE
City College of Calamba
2022 - 2026

ACCOUNTANCY, BUSINESS AND MANAGEMENT
Saint Benilde International School
2020 - 2022

JUNIOR HIGH SCHOOL
Sta. Cecilia Catholic School Inc.
2016 - 2020

ELEMENTARY
Mayapa Elementary School
2010 - 2016

Skills

- Time Management
- Research Writing
- Canva
- Teamwork and Collaboration
- Ready to learn and try innovative ways of improving work
- Attentiveness
- Creativity
- Microsoft Office

Language

Filipino
English

OBJECTIVE

Motivated and adaptable individual seeking an opportunity to contribute my skills while gaining professional experience in a dynamic organization. Committed to delivering quality work, learning continuously, and supporting team goals.

Work Experience

2022 -
2023

McDonald's Checkpoint

Service Crew

- I learn how to communicate to others, and I know how to handle difficult attitude's of people. Also, I learn how to boost my skills in public speaking.

2025

Day One's Cafe

Barista/kitchen (part time)

- Multiskilled café staff with experience as a barista and kitchen aide, skilled in beverage preparation, food handling, and customer service.

References

Kaye Arcilla
McDonalds / Manager
Phone: 09651489749
Email: arcillakayed@gmail.com

Gizelle Albaira
Day One Cafe / Owner
Phone: 09279986214
Email: gizellealbaira@gmail.com


ANN PAULINA P. CALIDA
Applicant's Signature



CITY COLLEGE OF CALAMBA

DEPARTMENT OF COMPUTING AND INFORMATICS
Bachelor of Science in Computer Science

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LEMUEL G. DICHE

 Calamba City, Laguna
 (+63) 9615280910
 lemuelgonzales0089@gmail.com



EDUCATIONAL ATTAINMENT

TERTIARY	CITY COLLEGE OF CALAMBA Bachelor of Science in Computer Science 2022 - Present
SECONDARY	PUNTA INTEGRATED SCHOOL Academic Track - Humanities and Social Sciences 2020 - 2022 PUNTA INTEGRATED SCHOOL 2016 - 2020
PRIMARY	BARANDAL ELEMENTARY SCHOOL 2010 - 2016

WORK EXPERIENCE

Company :Shopee
Position :Production Operator
Period of Employment :June-October 2025

SKILLS

- Software Knowledge: Basic Computer Skills
- Soft Skills: Responsibility, Adaptability, Willingness to learn

PERSONAL INFORMARTION

Age :21 years old
Date of Birth :November 24, 2004
Place of Birth :Catanauan, Quezon
Civil Status :Single
Citizenship :Filipino
Religion :Roman Catholic